



# The Cognitive Resonance of Diagrams

Gabriel Greenberg, UCLA • 10.2.26  
ELM 3, UConn

1. Diagrams and diagram systems
2. Why do some diagram systems flourish?
3. Cognitive resonance
4. Visual strategies

# Diagrams & Diagram Systems



**ARBOR**

There are likenesses, or **icons**; which serve to convey ideas of the things they represent simply by imitating them... There are **symbols**, or general signs, which have become associated with their meanings by usage.

— Peirce 1894 “What is a Sign?”



**ARBOR**

*The linguistic sign is arbitrary...* I mean that it is unmotivated, i.e. arbitrary in that it actually has no natural connection with the signified.

— Saussure 1916 “Course in General Linguistics”

# Symbolic representation

Words

Sentences

Mathematical  
Formalism

Non-Linguistic  
Emblems

**ARBOR**

**EQUOS**

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der folgenreichsten  
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icht immer etwas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\neg p \rightarrow (p \rightarrow q)$$

$$\neg p \rightarrow [p \rightarrow (q \wedge \neg q)]$$

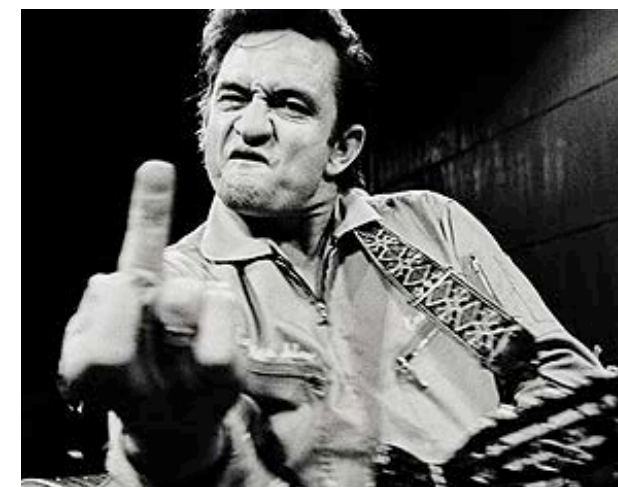
$$(p \rightarrow \neg p) \rightarrow [p \rightarrow (q \wedge \neg q)]$$

$$p \rightarrow (p \vee q)$$

$$p \rightarrow [p \vee (q \wedge \neg q)]$$

$$p \rightarrow [(p \wedge q) \vee (p \wedge \neg q)]$$

- (1)  $x = x$
- (2)  $\forall y (y \neq x) \rightarrow x \neq x$
- (3)  $x = x \rightarrow \neg \forall y (y \neq x)$
- (4)  $x = x \rightarrow \exists y (y = x)$
- (5)  $\exists y (y = x)$
- (6)  $\Box \exists y (y = x)$
- (7)  $\forall x \Box \exists y (y = x)$



# Iconic representation

3D Models



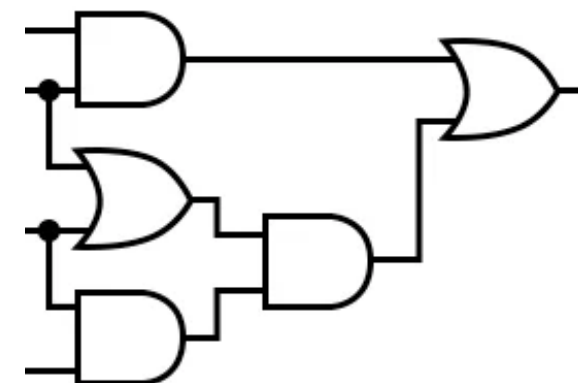
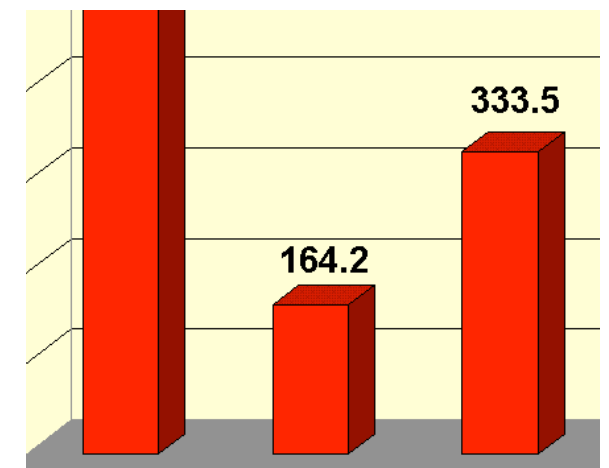
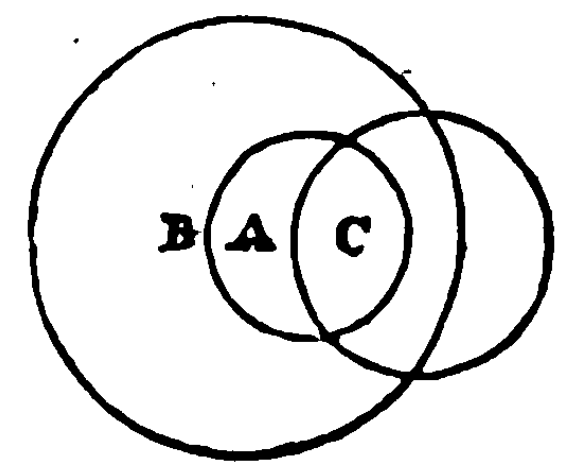
Pictures



Maps



Diagrams



## Symbolic representation

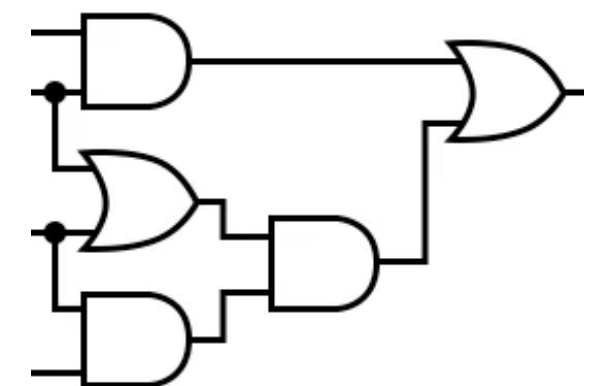
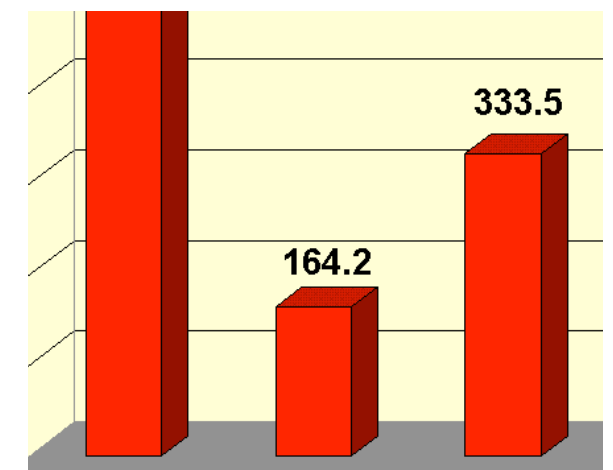
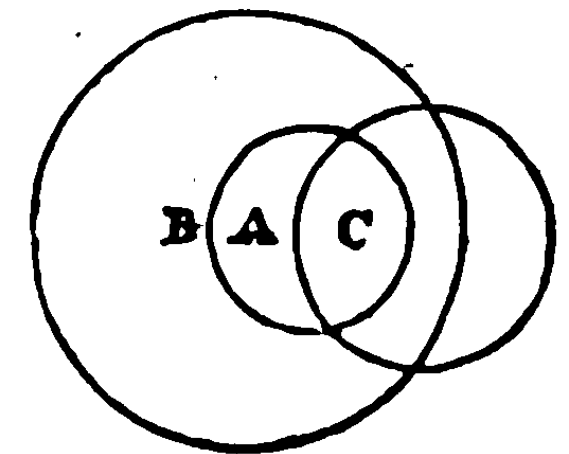
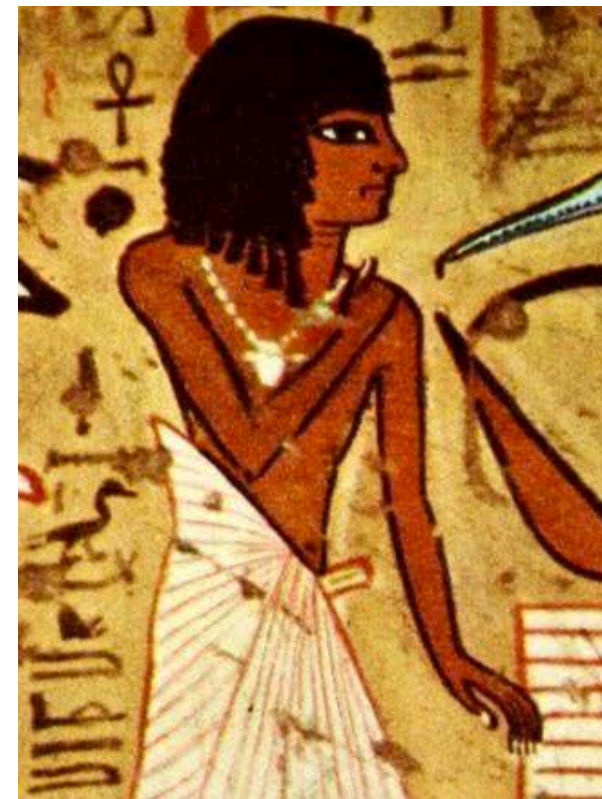
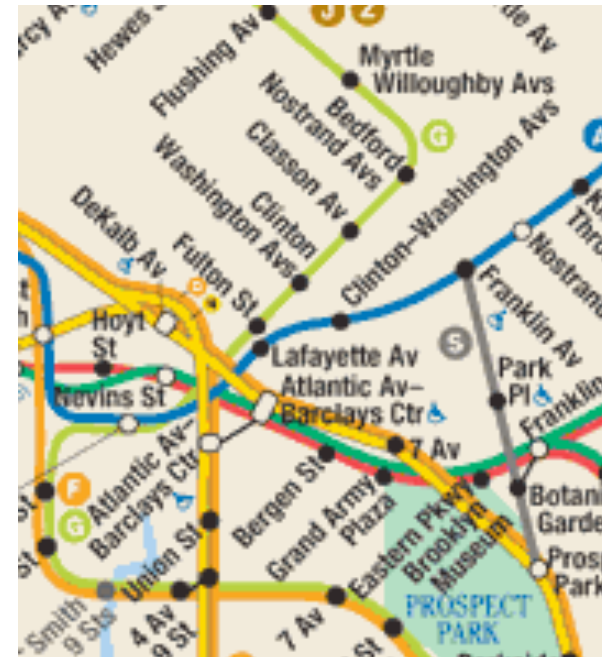
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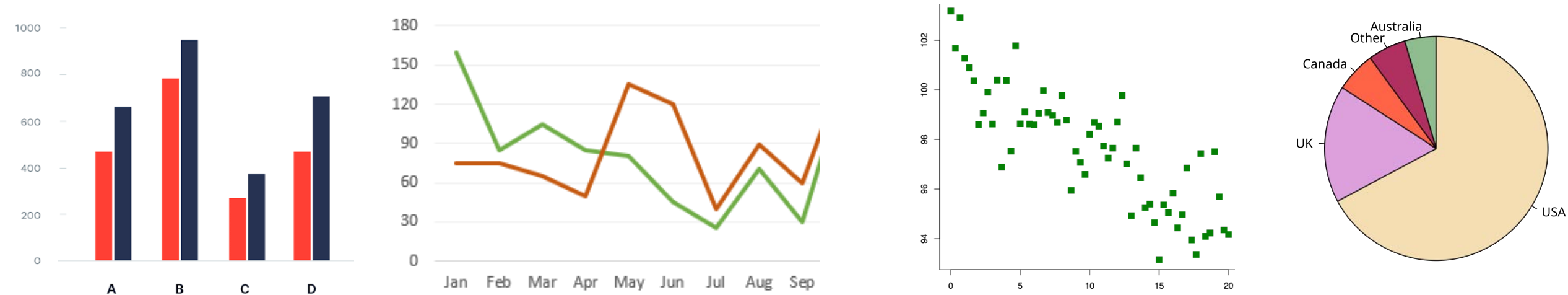
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## Iconic representation

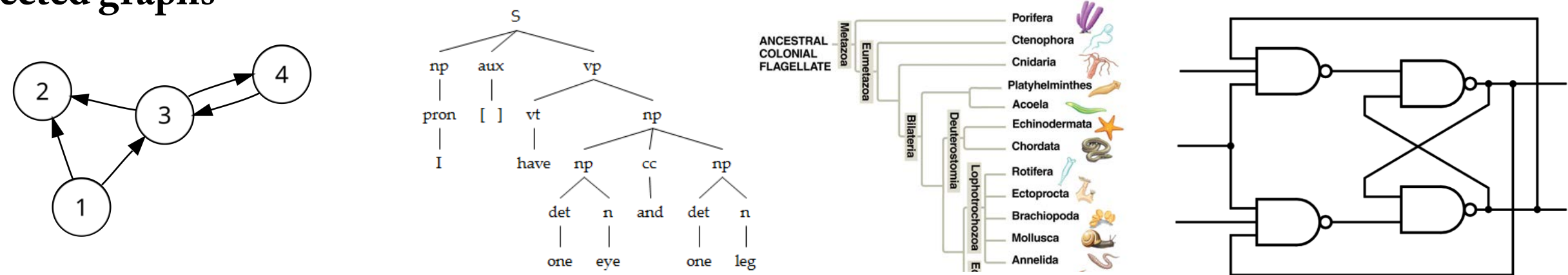


# Varieties of diagram

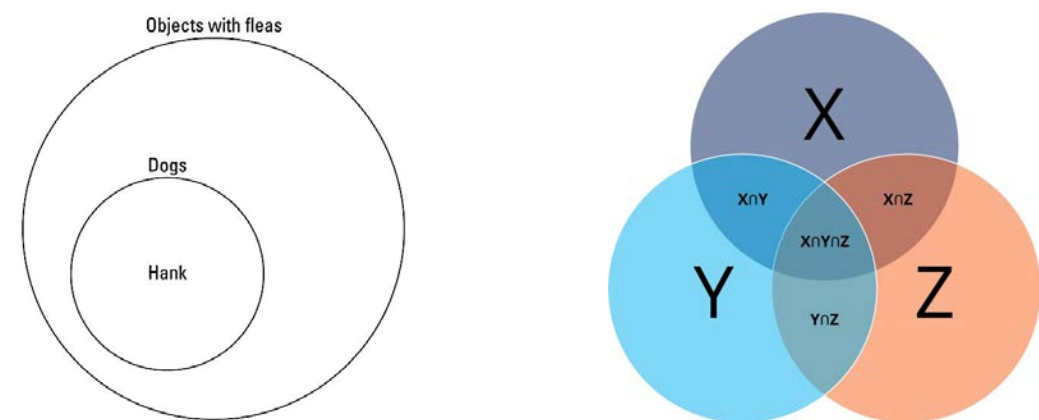
## Data charts



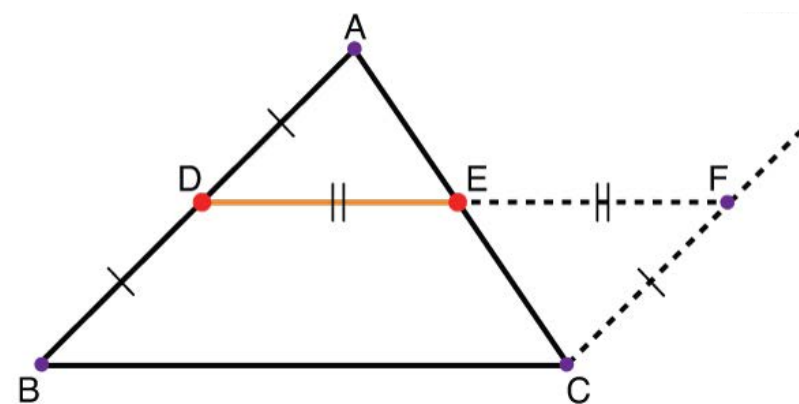
## Directed graphs



## Enclosure diagrams



## Geometric diagrams



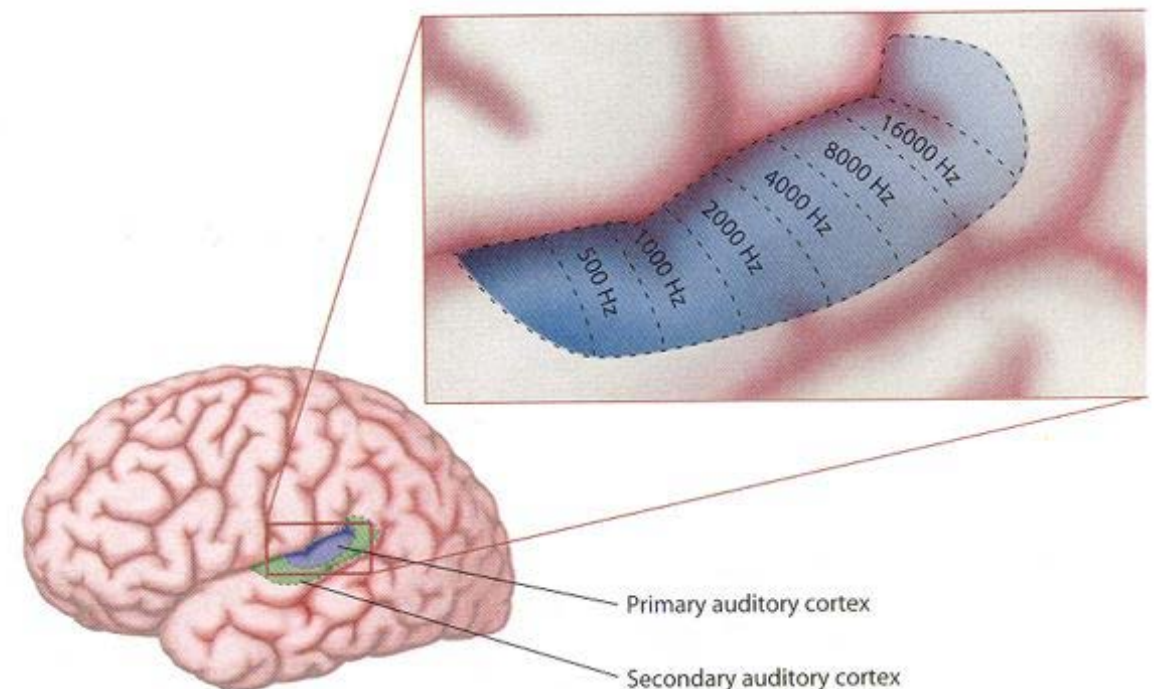
## Diagrams in face-to-face communication

- Gesture (diagrams with hands)
- Iconic sign language
- Facial expressions?
- Boardwork



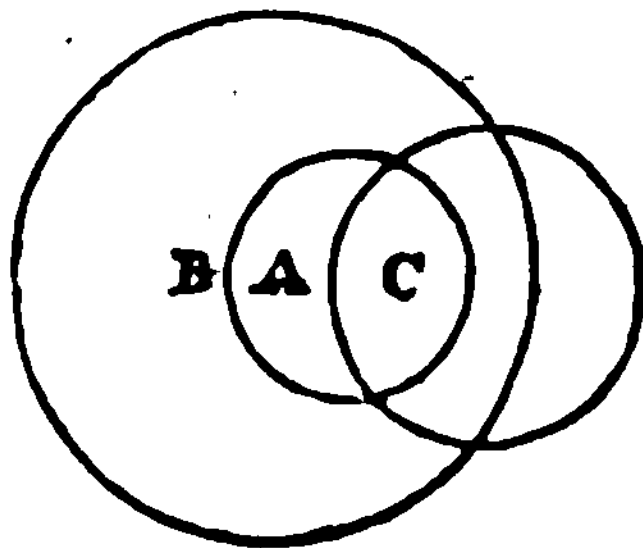
## Diagrams in the mind

- Analogue magnitude representations
- Tonotopic maps in A1
- Orientation maps in V1
- And many more...  
in core cognition and perception.



# Diagrams characterized

- Typical diagrams are signs that
  - Are composed iteratively from **simple graphical elements** and **parametric variation** of features. (Syntax)
  - Represent bodies of **relational data**. (Content)
  - Use aspects of **sign space** to represent positions in **feature space**. (Semantics)
  - Do *not* express content relative to a **spatial viewpoint**.



Euler diagram

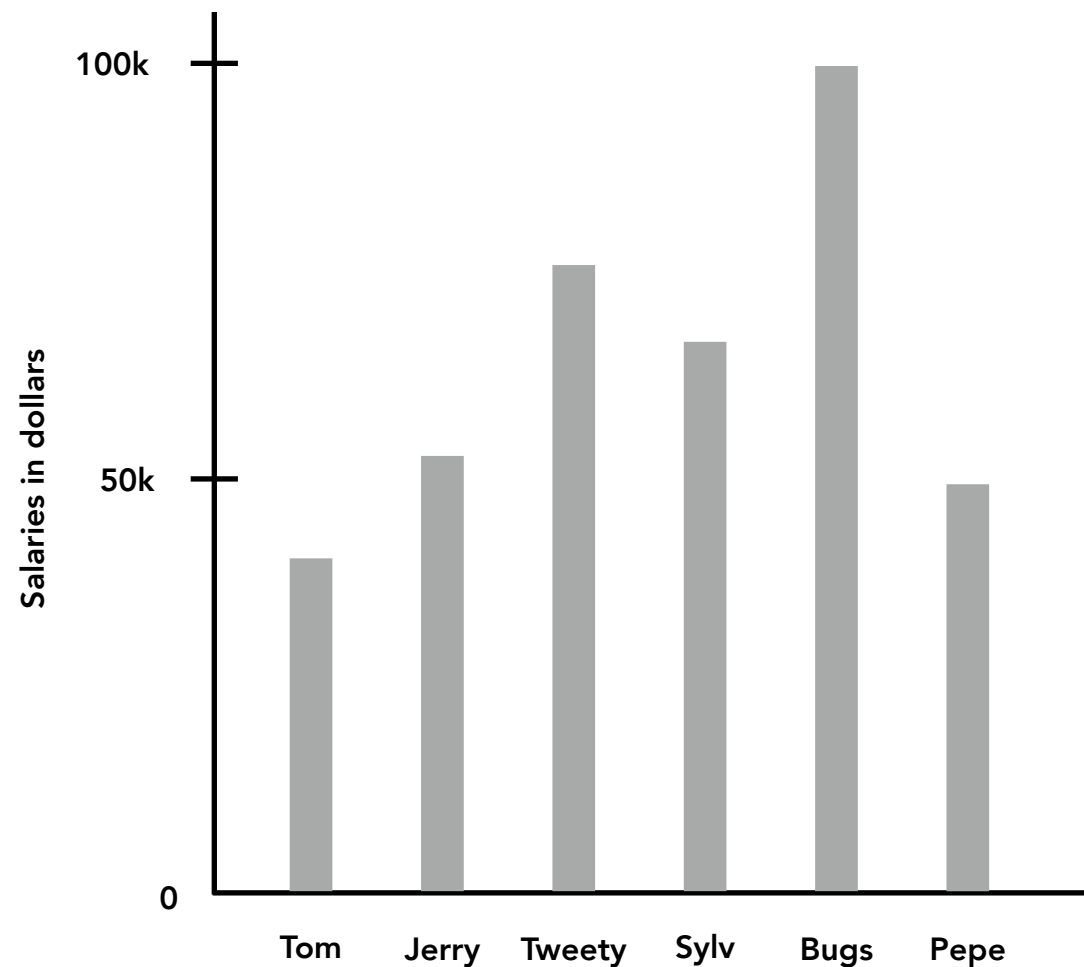


not a diagram

# Bar charts

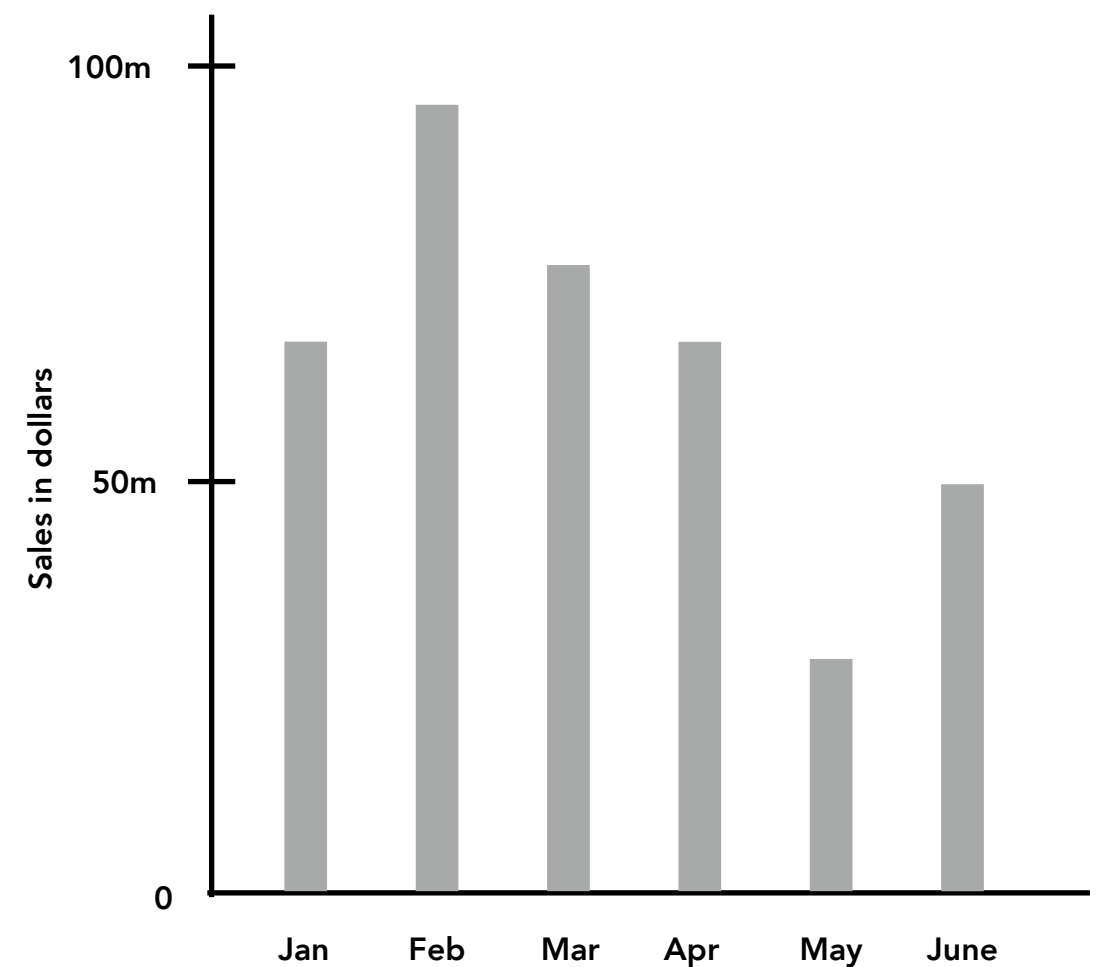
- **Bar charts** are diagrams for representing **two-dimensional data** using
  - bar **length** along the Y-axis to represent **continuous values** in the first data dimension,
  - **position** along the X-axis for **categorical or ordinal values** in the second data dimension.

Categorical bar chart



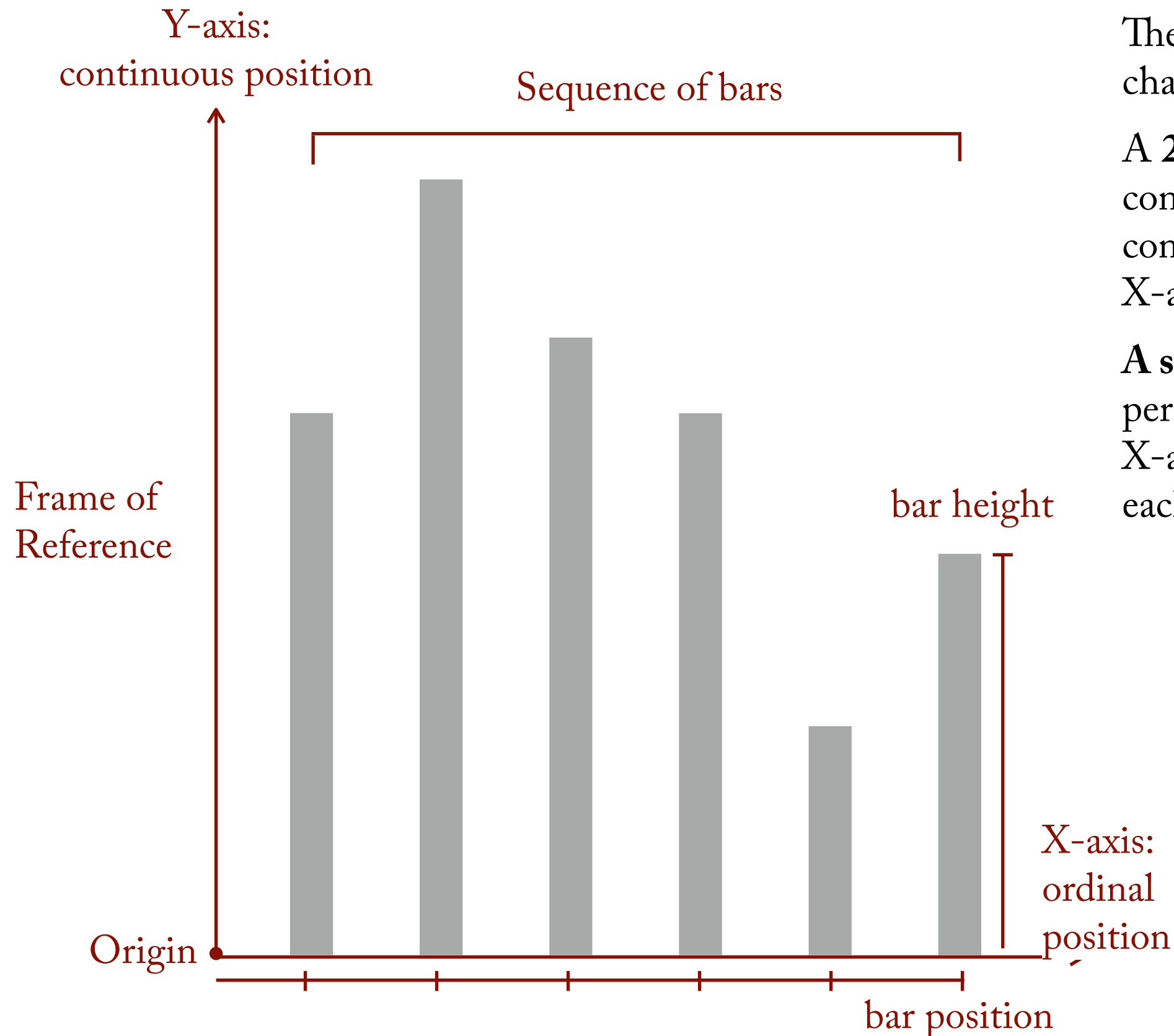
Employee salaries, ACME Corp., 2025

Ordinal bar chart



Sales by Month, ACME Corp., 2025

# Bar charts: syntax



The **syntax** of an ordinal bar chart is defined by:

A **2D frame of reference**, consisting of an origin, continuous Y-axis, and ordinal X-axis.

A **sequence of bars** that extend perpendicularly from the X-axis at variable heights, for each ordinal position.

## Bar charts: content

- The **content** of a bar chart is a state of affairs in which a **2-dimensional data set** holds.
- A 2-dimensional data set is defined by:
  - a set of elements  $E$ ;
  - two feature dimensions,  $S1$  and  $S2$ ;
  - where every element in  $E$  has one property from  $S1$  and one property from  $S2$ .
- For time-series graphs, the elements of  $E$  are typically *events*.

### Data Set 1: Salary/Employee (Categorical)

<e1, Tom, 40,000>  
<e2, Jerry, 53,000>  
<e3, Tweety, 76,000>  
<e4, Sylv, 67,000>  
<e5, Bugs, 100,000>  
<e6, Pepe, 50,000>

### Data Set 2: Sales/Month (Ordinal)

|                            |                       |
|----------------------------|-----------------------|
| <e1, January, 67,000,000>  | <e1, 1st, 67,000,000> |
| <e2, February, 95,000,000> | <e2, 2nd, 95,000,000> |
| <e3, March, 76,000,000>    | <e3, 3rd, 76,000,000> |
| <e4, April, 67,000,000>    | <e4, 4th, 67,000,000> |
| <e5, May, 28,000,000>      | <e5, 5th, 28,000,000> |
| <e6, June, 50,000,000>     | <e6, 6th, 50,000,000> |

## Bar charts: semantics

- Let  $D$  be an ordinal bar chart diagram for sales/month data.

$Content(D)$  = the state of affairs such that:

for every bar  $b$  in  $D$ :

there is an event  $e$ :

- (i)  $Y\text{-height}(b) \times k = sales(e)$ ;
- (ii)  $ev(X\text{-rank}(b), Months) = month(e)$ .

- $Y\text{-height}$  is a function from lines to the distance between their least  $y$ -coordinate and their greatest  $y$ -coordinate.
- $k$  is a system-defined constant.
- $X\text{-rank}$  is a function from lines to the ordinal rank along the  $X$ -axis of their least  $x$ -coordinate.
- $ev$  is a function from a positive integer  $n$  and a sequence, to the  $n$ th item in that sequence.

# Bar charts: semantics

$Content(D)$  = the state of affairs such that:

for every bar  $b$  in  $D$ :

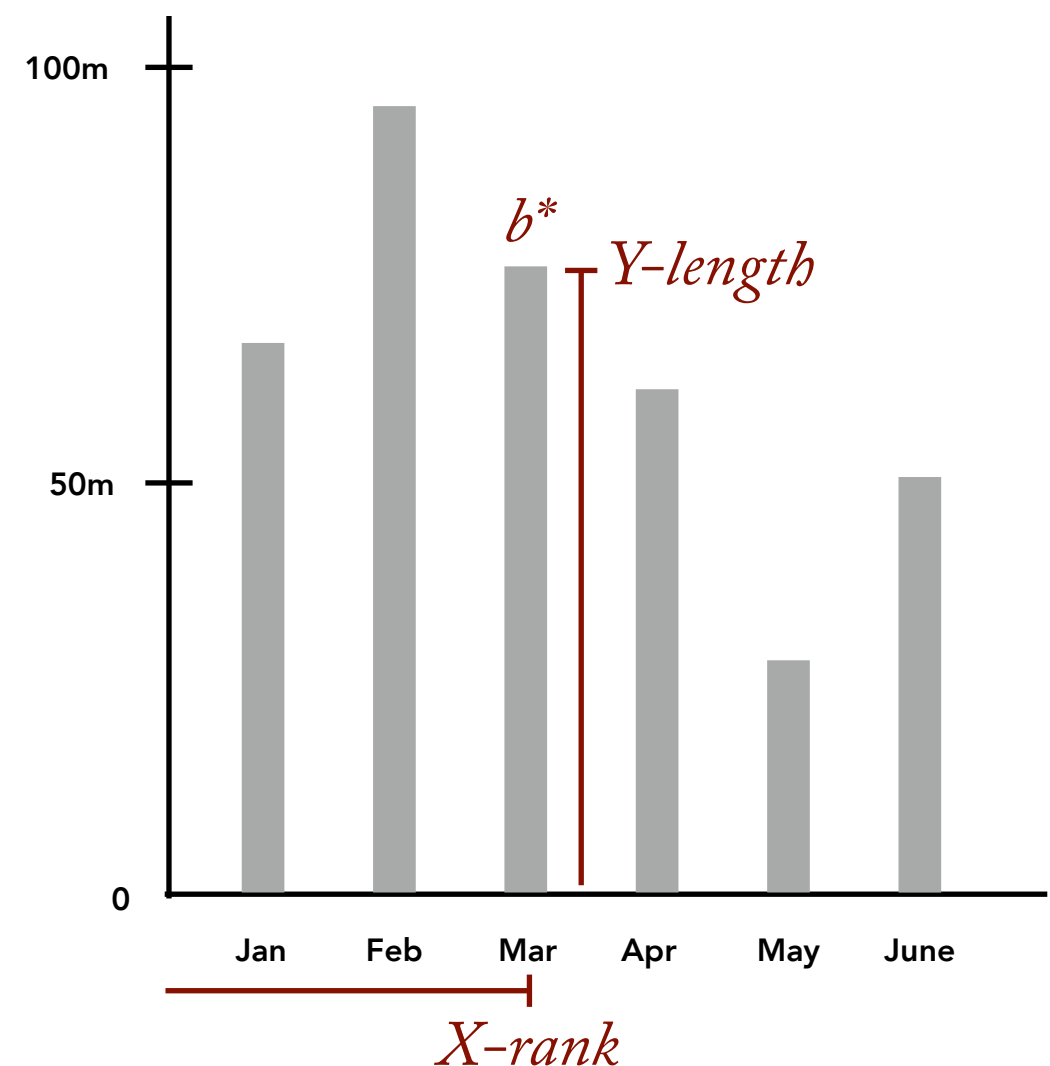
there is an event  $e$ :

- (i)  $Y\text{-length}(b) \times k = sales(e)$ ;
- (ii)  $ev(X\text{-rank}(b), Months) = month(e)$ .

Assume  $k = 156,000$ .

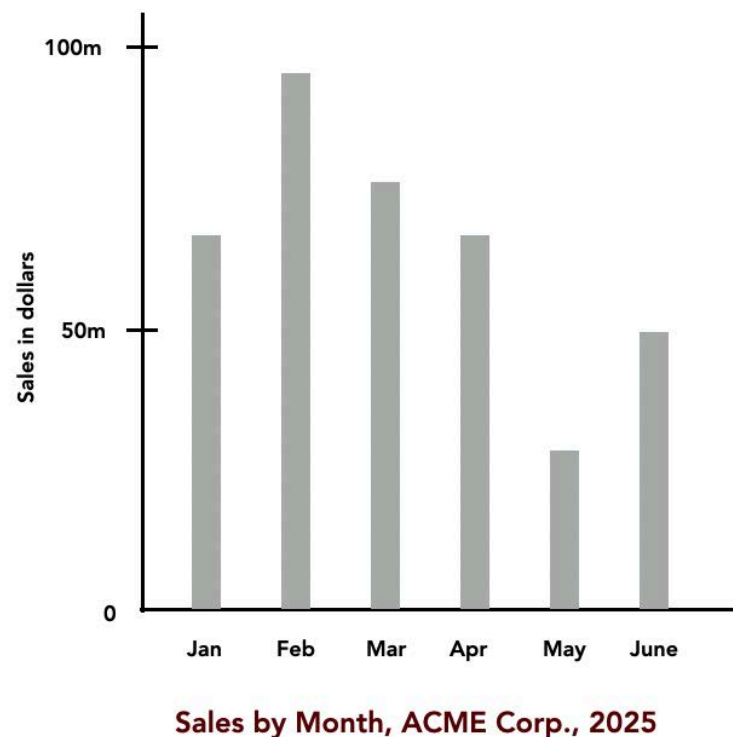
$Content(D)$  includes: there is an event  $e$ :

- (i)  $Y\text{-length}(b) \times k = sales(e) \Rightarrow$   
 $487 \times 156k = 76m = sales(e)$
- (ii)  $ev(X\text{-rank}(b), Months) = month(e) \Rightarrow$   
 $ev(3, Months) = March = month(e)$



# Why encode information diagrammatically?

- **Diagram semantics:** compressed, easy to learn, naturally extended, free-ride inference.
- **Linguistic semantics:** complex, memory intensive, limited by lexicon, explicit inference.



<e1, January, 67,000,000>

<e2, February, 95,000,000>

<e3, March, 76,000,000>

<e4, April, 67,000,000>

<e5, May, 28,000,000>

<e6, June, 50,000,000>

$$(i) \quad Y\text{-height}(b) \times k = \text{sales}(e);$$

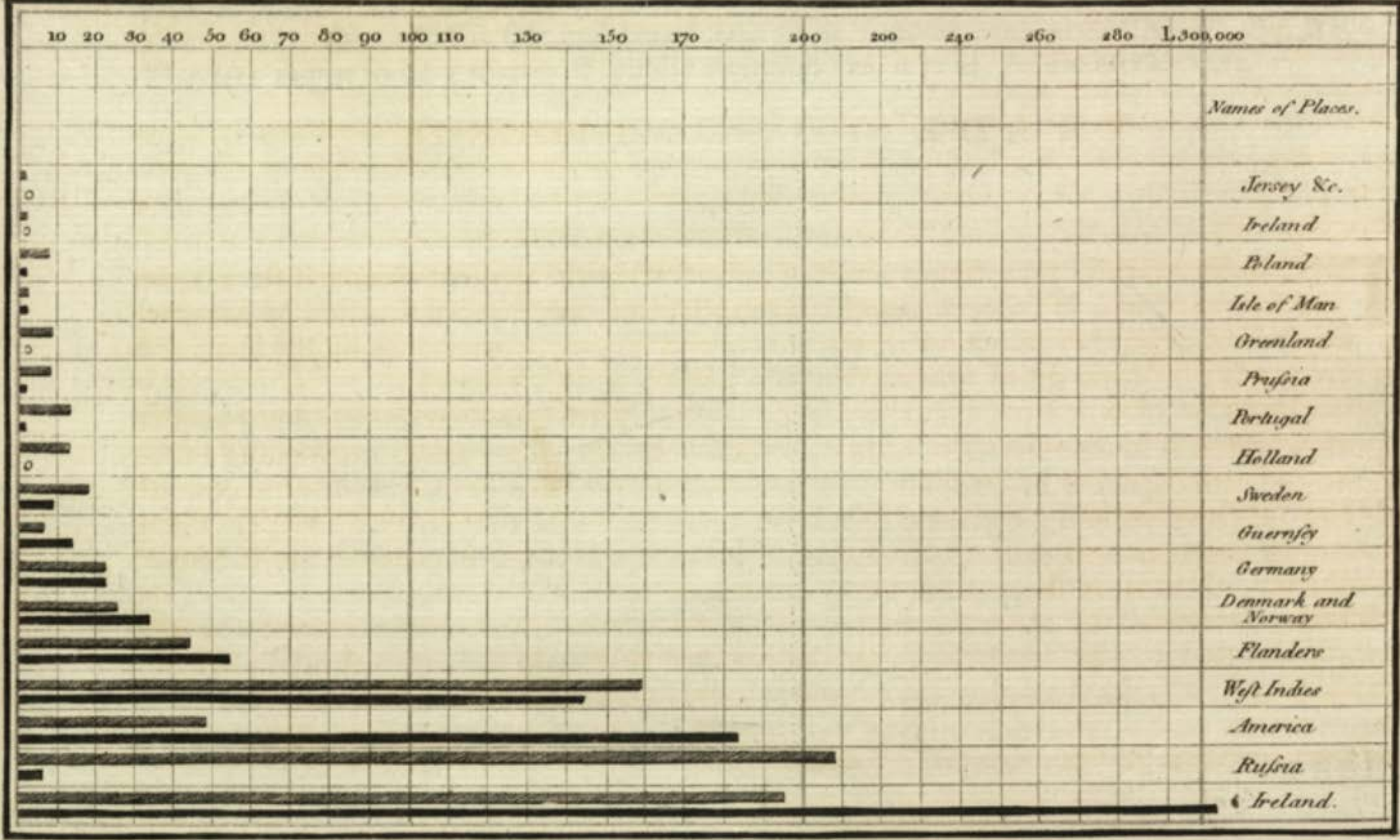
$$(ii) \quad \text{ev}(X\text{-rank}(b), \text{Months}) = \text{month}(e).$$

Why do some  
diagram systems flourish?

# The contingency of diagrams

- **History:** particular diagram systems have particular histories.
- **Common diagram systems were *recently invented*.**
  - Many diagram systems have a particular beginning point. Diagrammatic explosion at the end of the 18th century:
    - Euler Diagrams (1768, Euler)
    - Bar Charts (1786, Playfair)

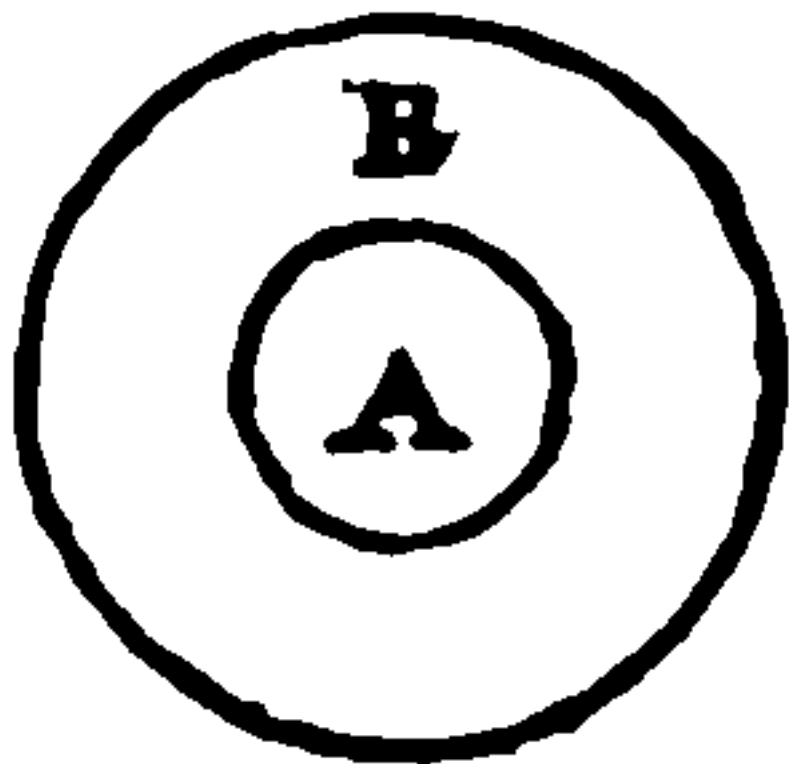
Exports and Imports of SCOTLAND to and from different parts for one Year from Christmas 1780 to Christmas 1781.



The Upright divisions are Ten Thousand Pounds each. The Black Lines are Exports the Ribbed lines Imports.

**The first bar graph** (more or less): William Playfair, 1786, “Exports and Imports of Scotland to and from different parts for one Year from Christmas 1780 to Christmas 1781”

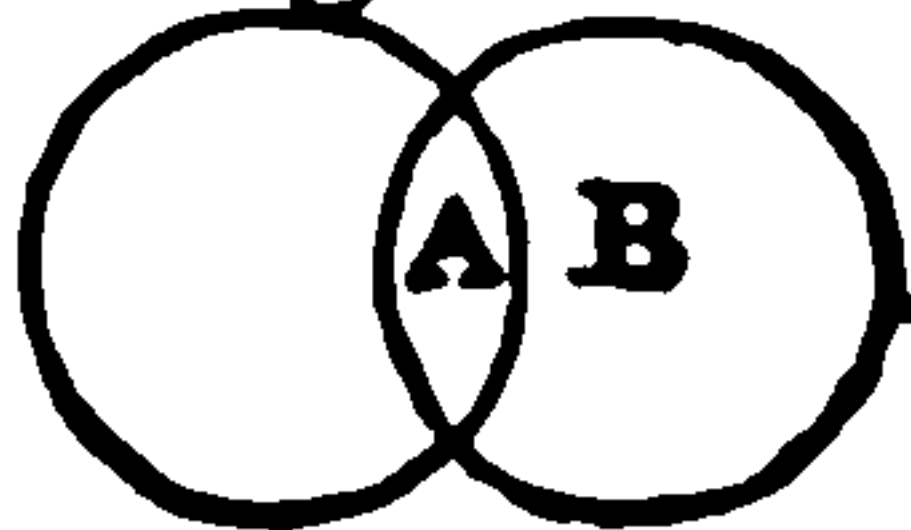
***Fig. 43.***



***Fig. 44.***



***Fig. 45.***



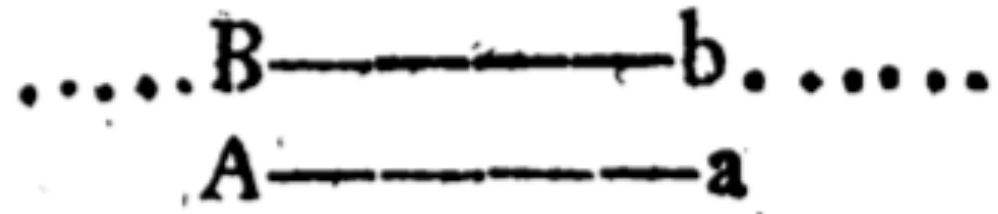
**The Euler Diagram.** Leonhard Euler, 1768, *Letters to a German Princess*.

## The contingency of diagrams

- **History:** particular diagram systems have particular histories.
- **Common diagram systems were *recently invented*.**
  - Many diagram systems have a particular beginning point. Diagrammatic explosion at the end of the 18th century:
    - Euler Diagrams (1768, Euler)
    - Bar Charts (1786, Playfair)
- **Cognitive basis.** Diagram comprehension for these systems cannot be part of our innate cognitive endowment.
- **Contrast with language and pictorial representation:**
  - (Most) **languages are ancient**, with no definite starting point. Conjectured starting point: ~150,000 yrs ago.
  - **Pictures and maps are ancient.** Earliest known picture: ~50,000 years ago. Earliest known maps: ~9,000 years ago.
  - Both are rooted in **innate cognitive architecture**.

## Why do (some) diagrams flourish?

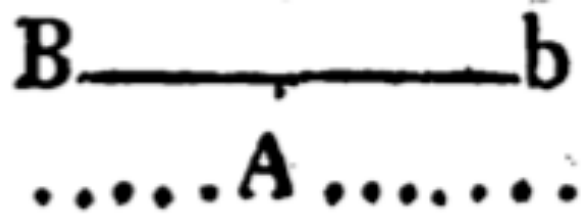
- Why do diagram systems **flourish**? Most don't.
- The ones we have today are the **all-stars** of cultural evolution.
  - Euler Diagrams (1768, Euler)
  - **Bar Charts** (1786, Playfair)
- Many common diagram systems were **recently invented**.
- But **why** did it take thousands of years (after the advent of writing) to invent a diagram system, whose syntax and semantics could be written down on a napkin?



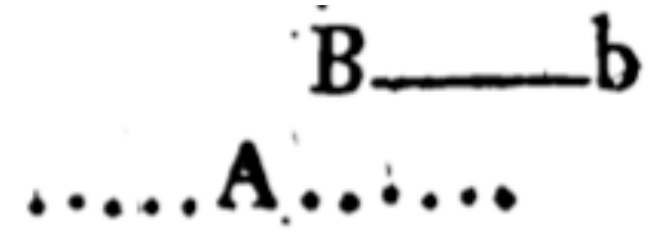
All As are Bs.



No A is B.



Some A is B.



Some A is not B.

**Lambert's Linear Diagrams.** Johann Heinrich Lambert, 1764, *Neues Organon*.

# What makes a diagram system successful?

- Three ways of asking this question:
- **Flourishing:** why do some diagram systems come into widespread global usage and endure, while others fade away?
- **Individual choice:** what factors drive an agent to select a given diagram system for use in a given context?
- **Invention:** what considerations drove the inventors of particular diagram systems to construct their systems as they did?
  - **Relatedly:** are there possible diagram systems that we don't yet use?  
Are our systems really the best possible systems?
- **Into the mind.** It seems the only way to answer these questions is to understand what makes diagrams *work* in the cognitive economy of their users.

## How do (some) diagrams work?

- What **cognitive capacities** are enlisted when we read diagrams?
- **Compare:**
  - **Natural language** leverages recursive, hierarchical, and compositional capacities for thought.
  - **Pictorial representation** exploits the visual system's capacity for the construction of perspective visual space.
  - What cognitive capacities make diagrams work?
- **The key: most diagram systems *don't* work** (well).

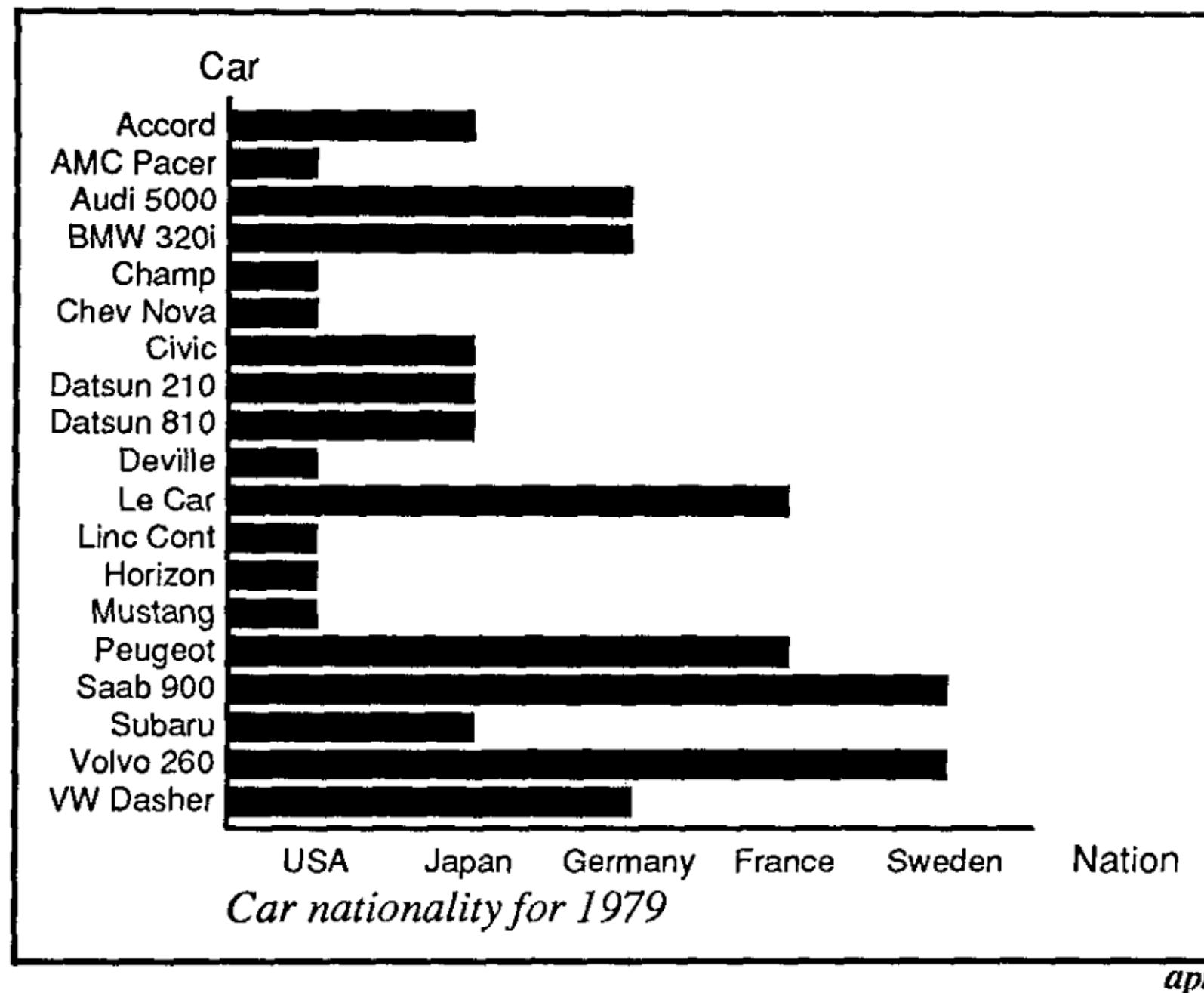
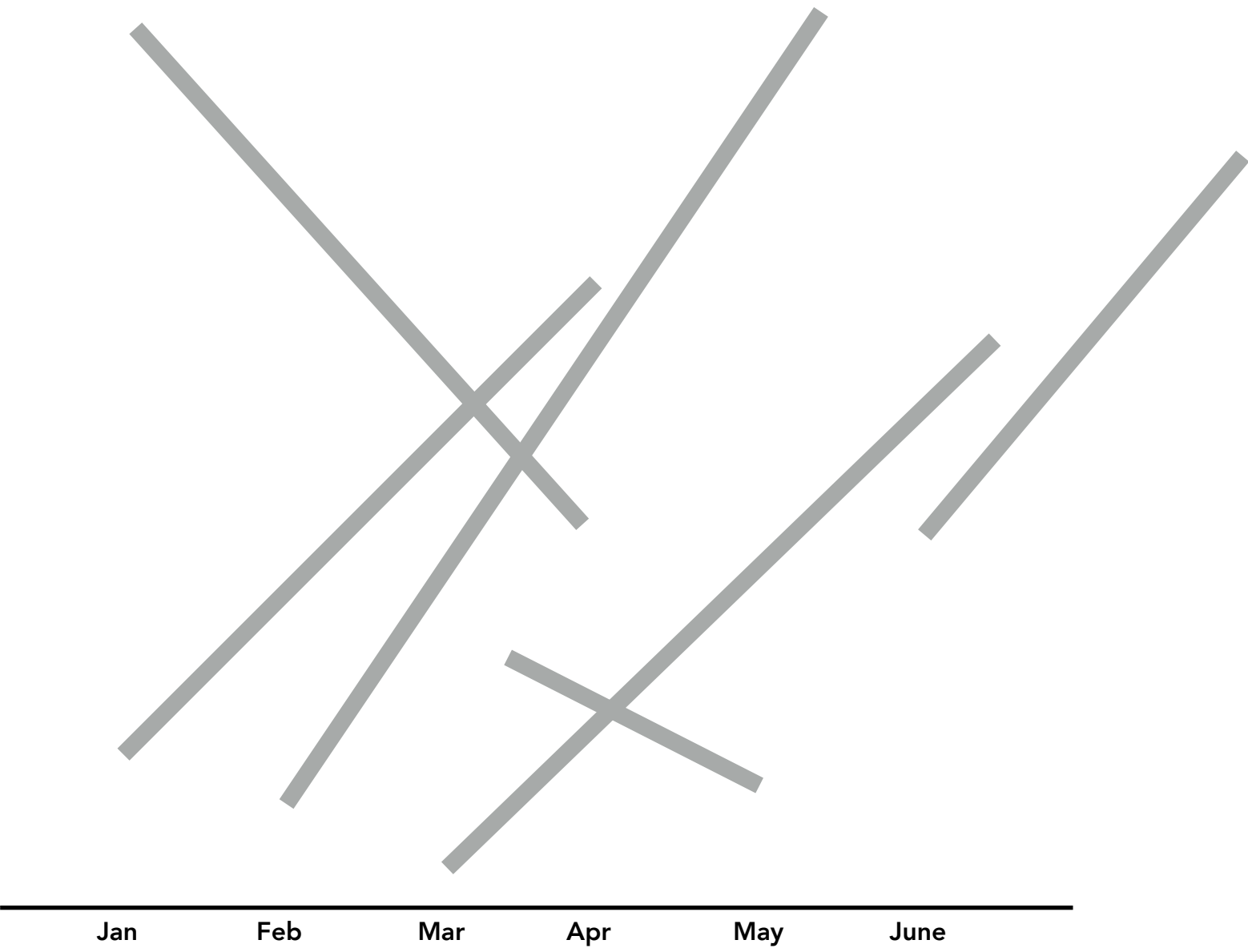


Fig. 11. Incorrect use of a bar chart for the *Nation* relation. The lengths of the bars suggest an ordering on the vertical axis, as if the USA cars were longer or better than the other cars, which is not true for the *Nation* relation.

**An (intentionally) bad diagram.** Jock Mackinlay, 1986, "Automating the design of graphical presentations of relational information"

# Dissonant diagrams, Case 1: scattered bars

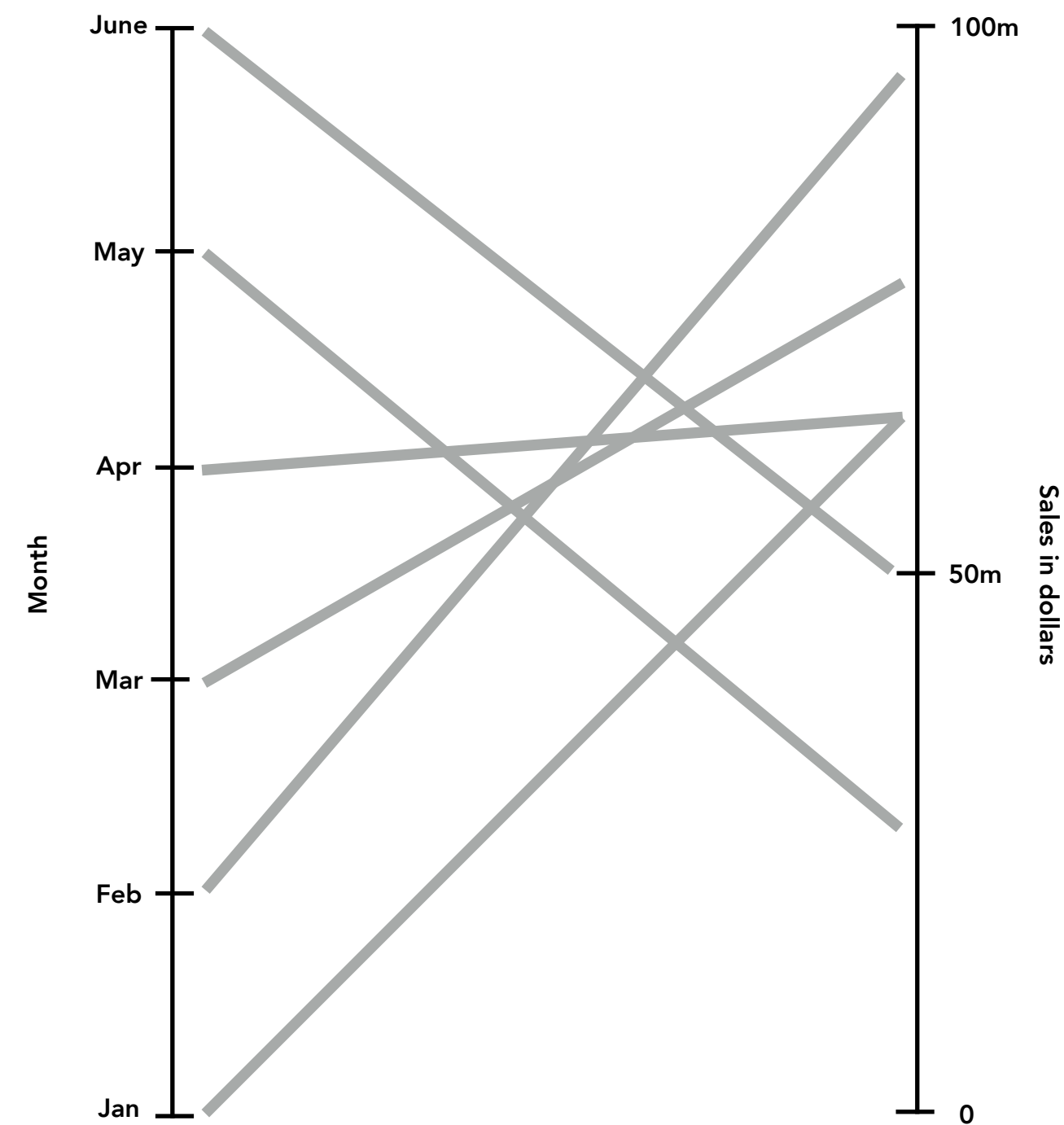


**Month** is represented by the ordinal rank of the x-position of the **bottom of the bar**.

**Sales** are represented by the length of the bar.

Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 2: slope graph



Sales by Month, ACME Corp., 2025

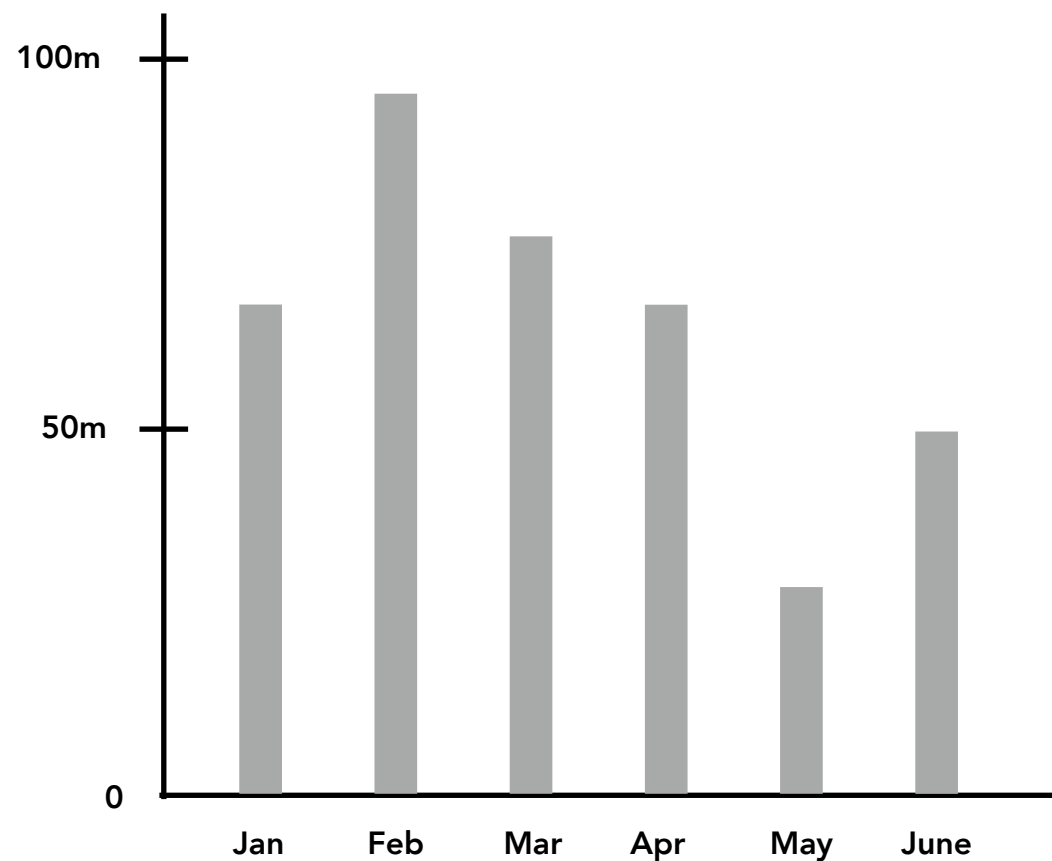
## Good systems, bad systems

- Most possible diagram systems don't facilitate cognitive engagement at all. The systems that flourished are a small, distinguished subset.
- **So the question becomes**, what distinguishes the diagram systems that do work from those that don't?
  - Iconicity, isomorphism, and other formal-semantic properties do not distinguish between good and bad diagram systems.
  - Why do some iconic systems work better than others?
- In search of a theory of **cognitive resonance**.

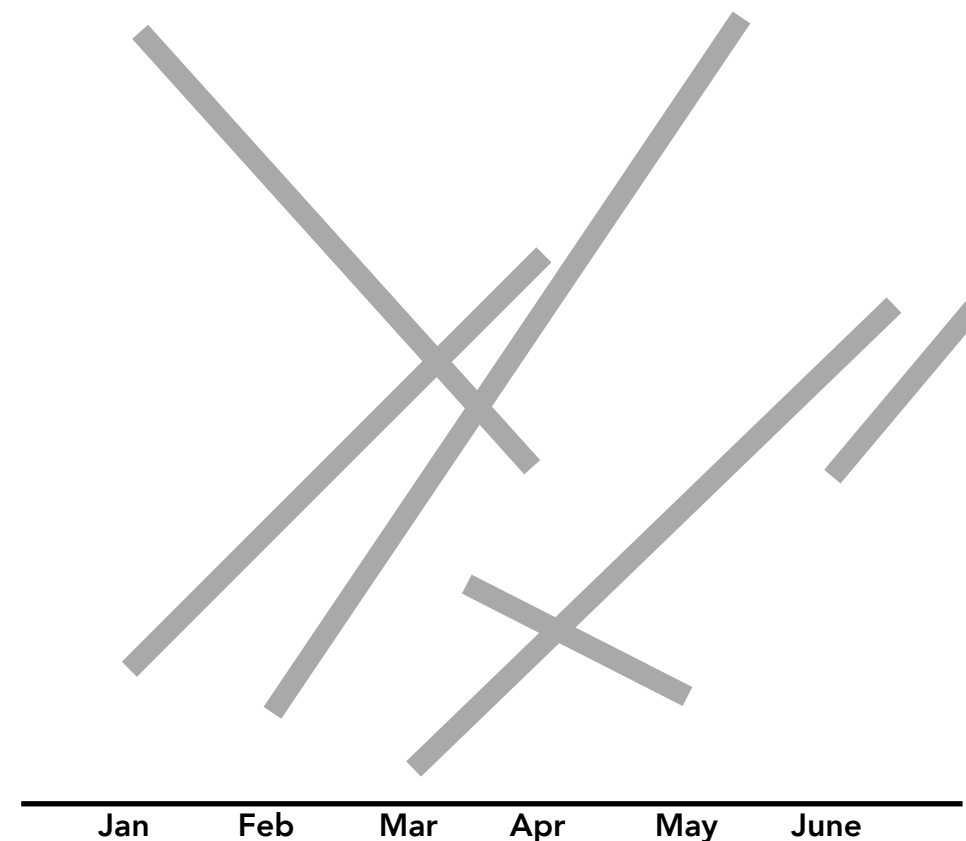
Cognitive resonance

# Cognitive resonance

- A diagram system is **cognitively resonant** for a population when that system flourishes in that population, in part due to the cognitive makeup of the members of the population.



Sales by Month, ACME Corp., 2025



Sales by Month, ACME Corp., 2025

Hypotheses:      cognitively resonant

cognitively dissonant

# Expressiveness and Effectiveness

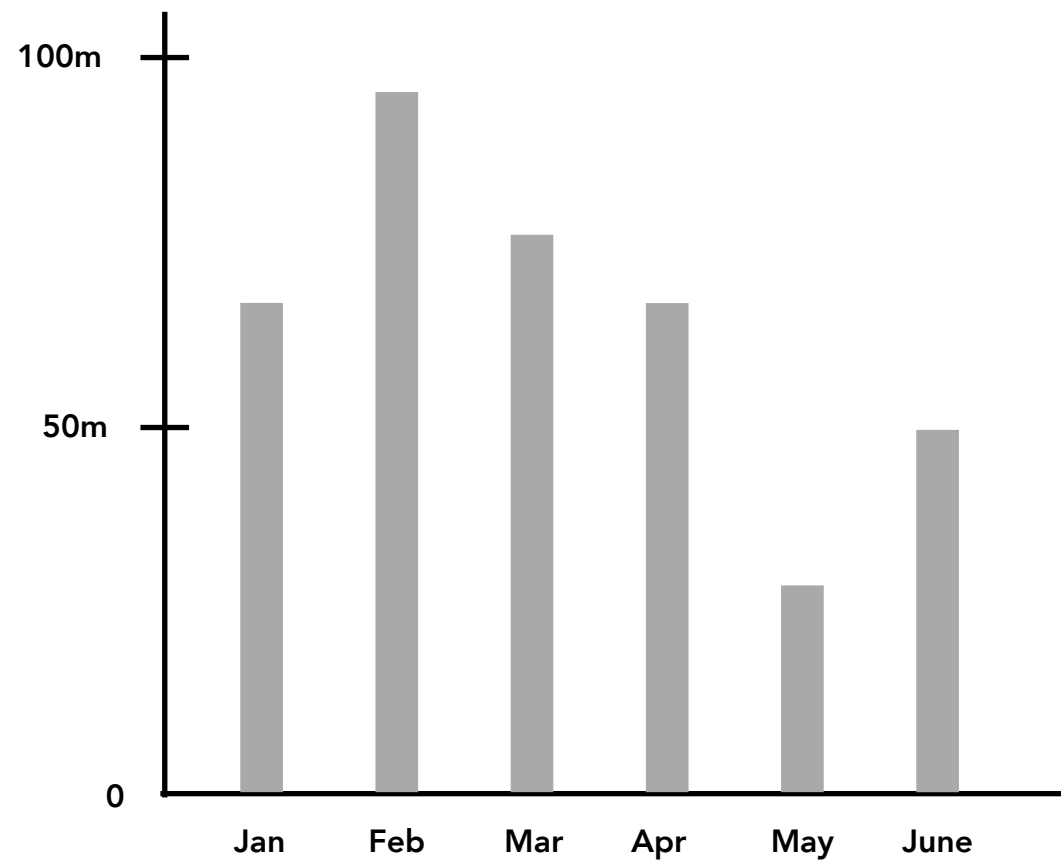
- **Expressiveness + Effectiveness** (Mackinlay 1986)
  - **Expressive adequacy:** does a diagram express all and only the target data?
  - **Effectiveness:** how well does a diagram exploit the interaction between the medium and the human cognitive system.
- **Effectiveness and aesthetic excellence.** Much literature on effectiveness targets excellence in diagram construction (Tufte 1983)
- **Soft and hard factors:**
  - **Soft factors:** visual salience, grouping, discriminability (graphic design).
  - **Hard factors:** does the architecture of a diagram system facilitate human use?

## Cognitive resonance vs. effectiveness

- A diagram system is **cognitively resonant** for a population when that system flourishes in that population, in part due to the cognitive makeup of the members of the population.
- **Diagrams and diagram systems**
  - Effectiveness is principally a property of **token diagrams**, including their graphic design.
  - Cognitive resonance is a property of **diagram systems**, independent of their graphic design.
- **Soft and hard factors**
  - Effectiveness includes both soft (graphic design) and hard (system design) factors.
  - Cognitive resonance targets hard factors at the architectural level.

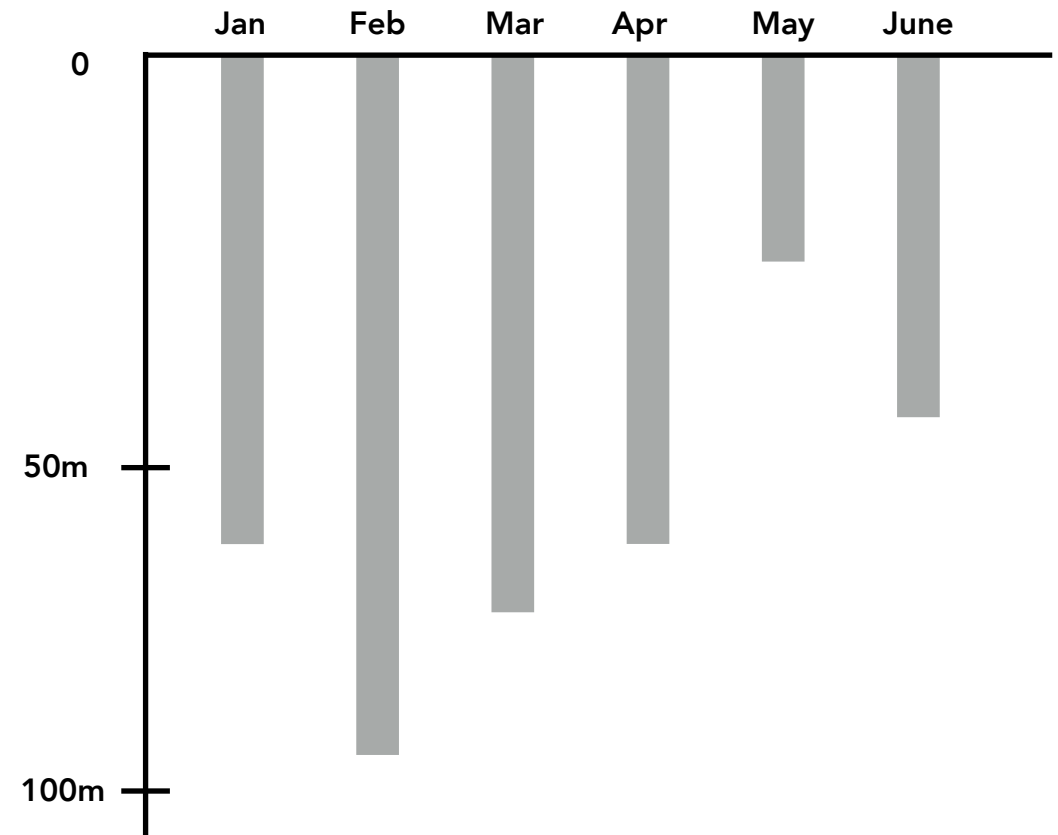
## Tversky's concept of congruence

- Tversky hypothesizes that **spatial schemas** drive diagram comprehension.
  - E.g. up/down, horizontal/vertical, center/surround, proximity/distance.
- There are values and expectations associated with each.
  - E.g., in the up/down schema, higher positions on the page are associated with growth, accumulation, and magnitude.
- A diagram is more or less **congruent** to the extent that it fits with the values and expectations of the spatial schemas it engages.



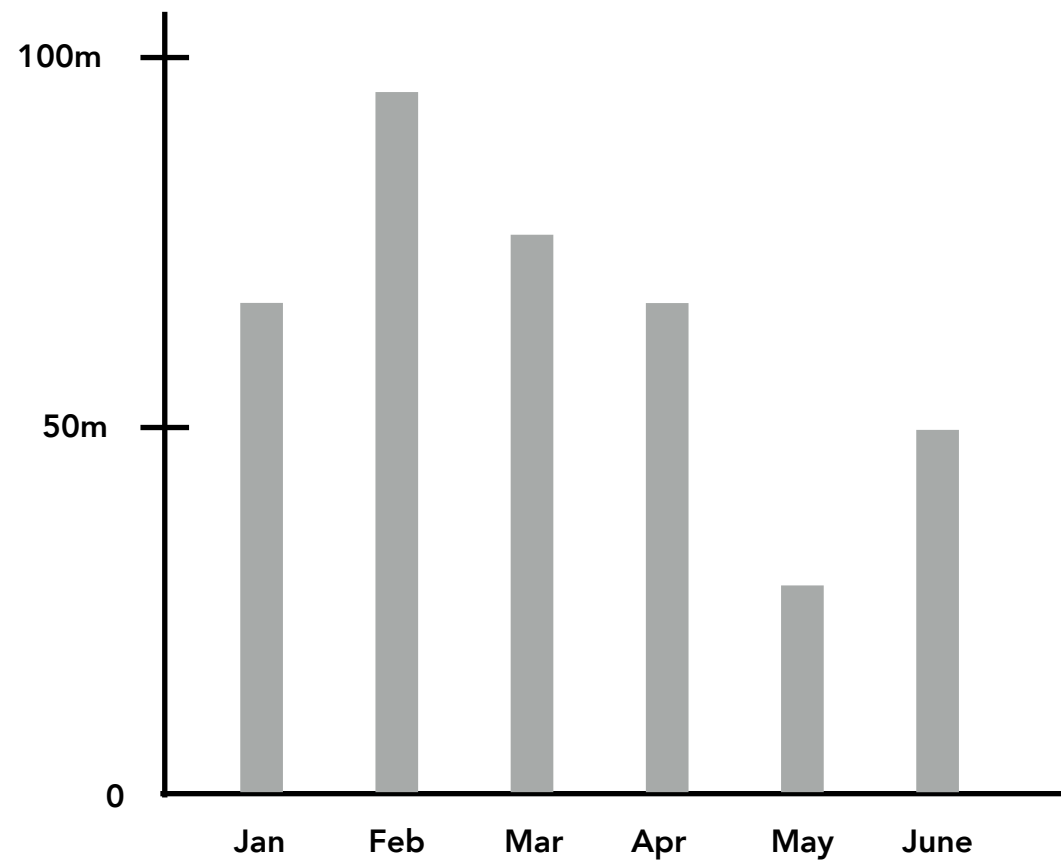
Sales by Month, ACME Corp., 2025

cognitively resonant



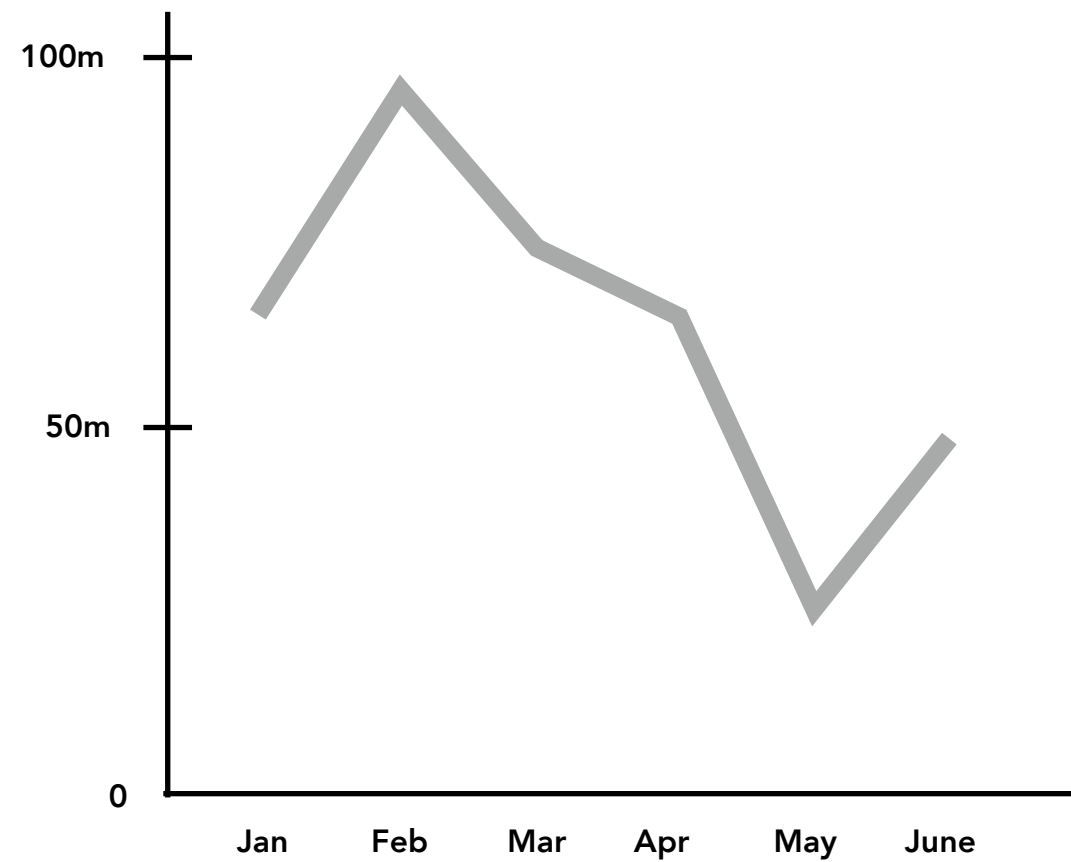
Sales by Month, ACME Corp., 2025

cognitively dissonant?



Sales by Month, ACME Corp., 2025

data-point oriented



Sales by Month, ACME Corp., 2025

trend oriented

## Mackinlay's scales

- Mackinlay developed a set of **primitive representational scales**. They describe what visual elements best represent different data structures.

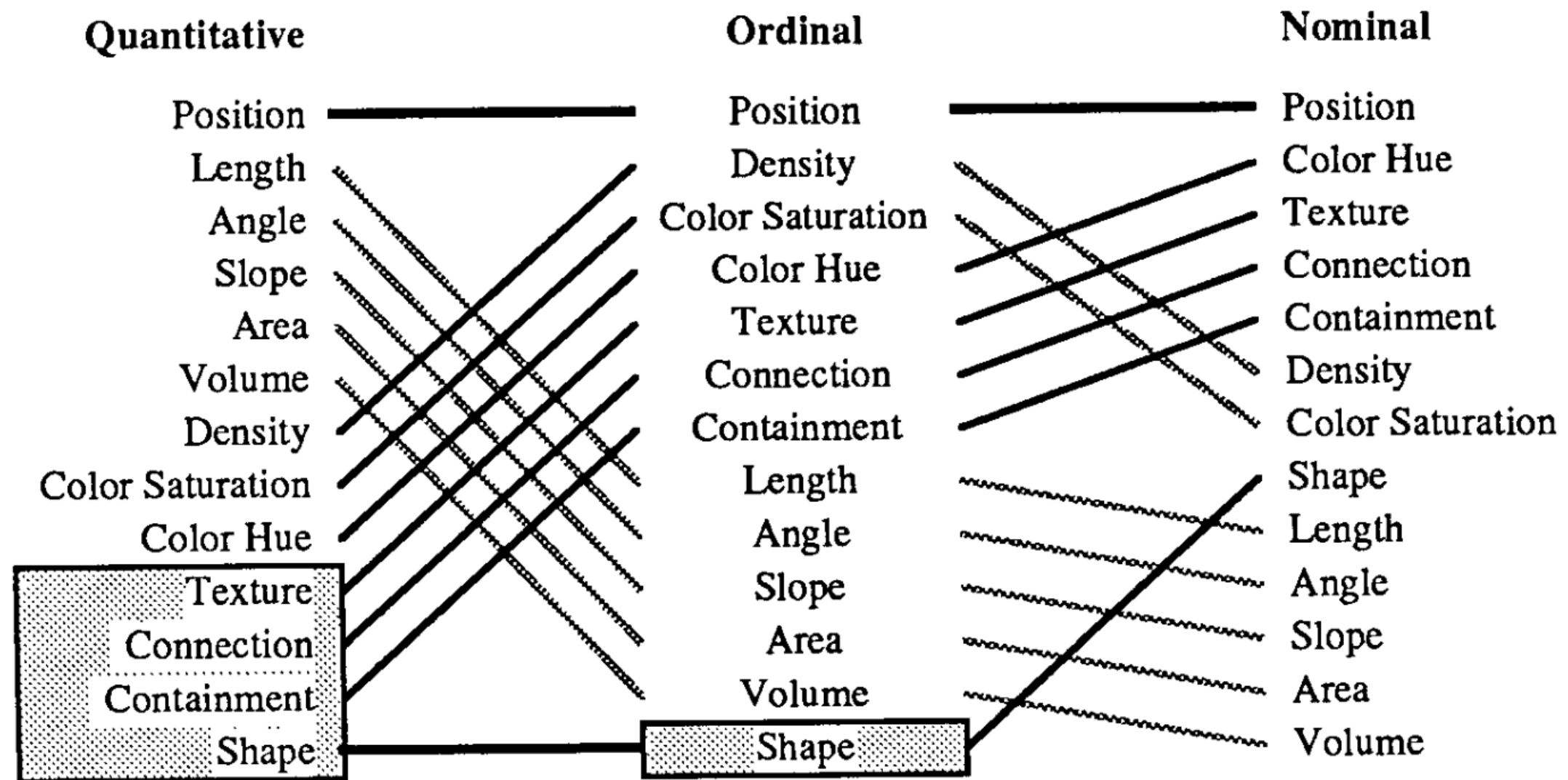


Fig. 15. Ranking of perceptual tasks. The tasks shown in the gray boxes are not relevant to these types of data.

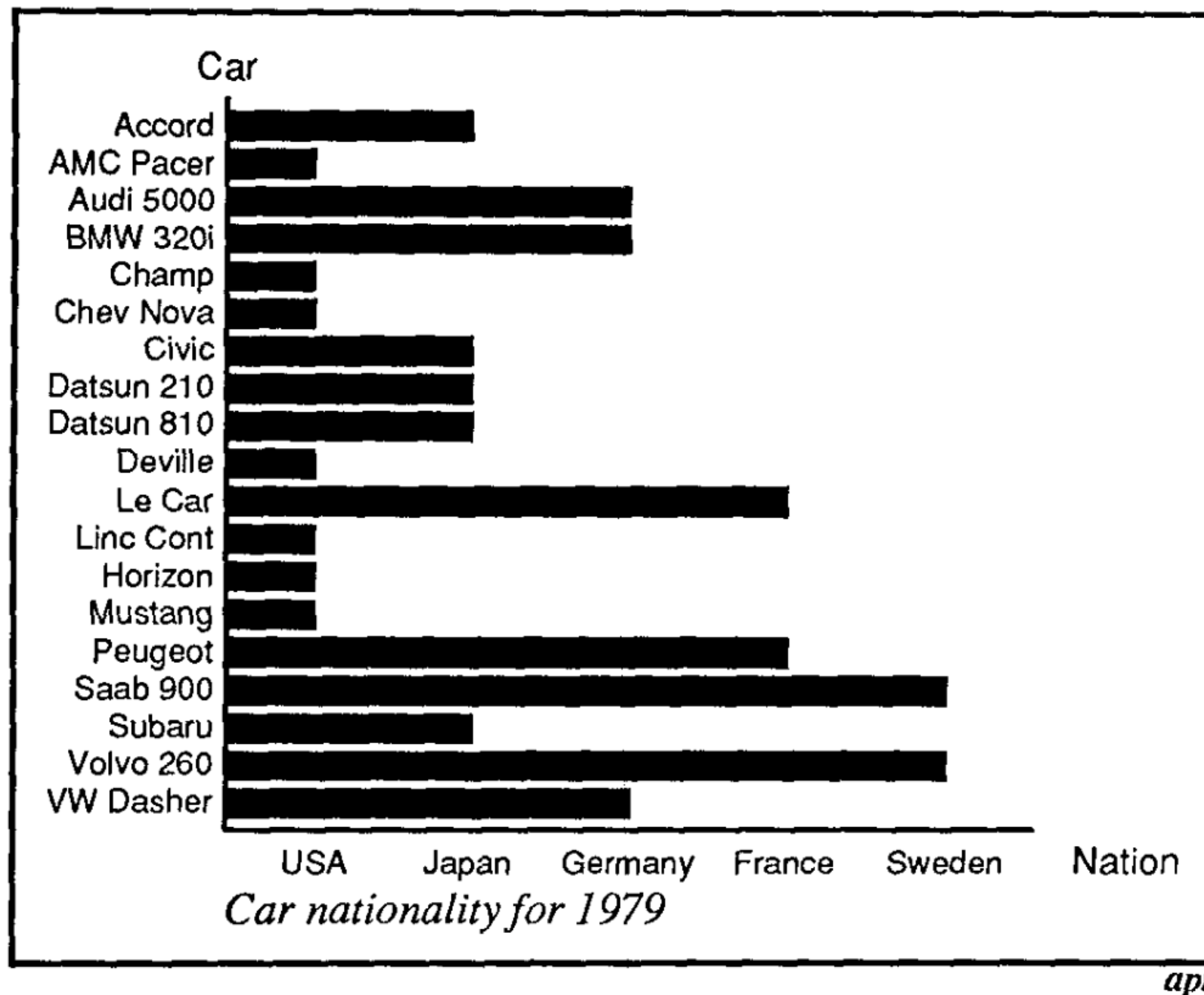
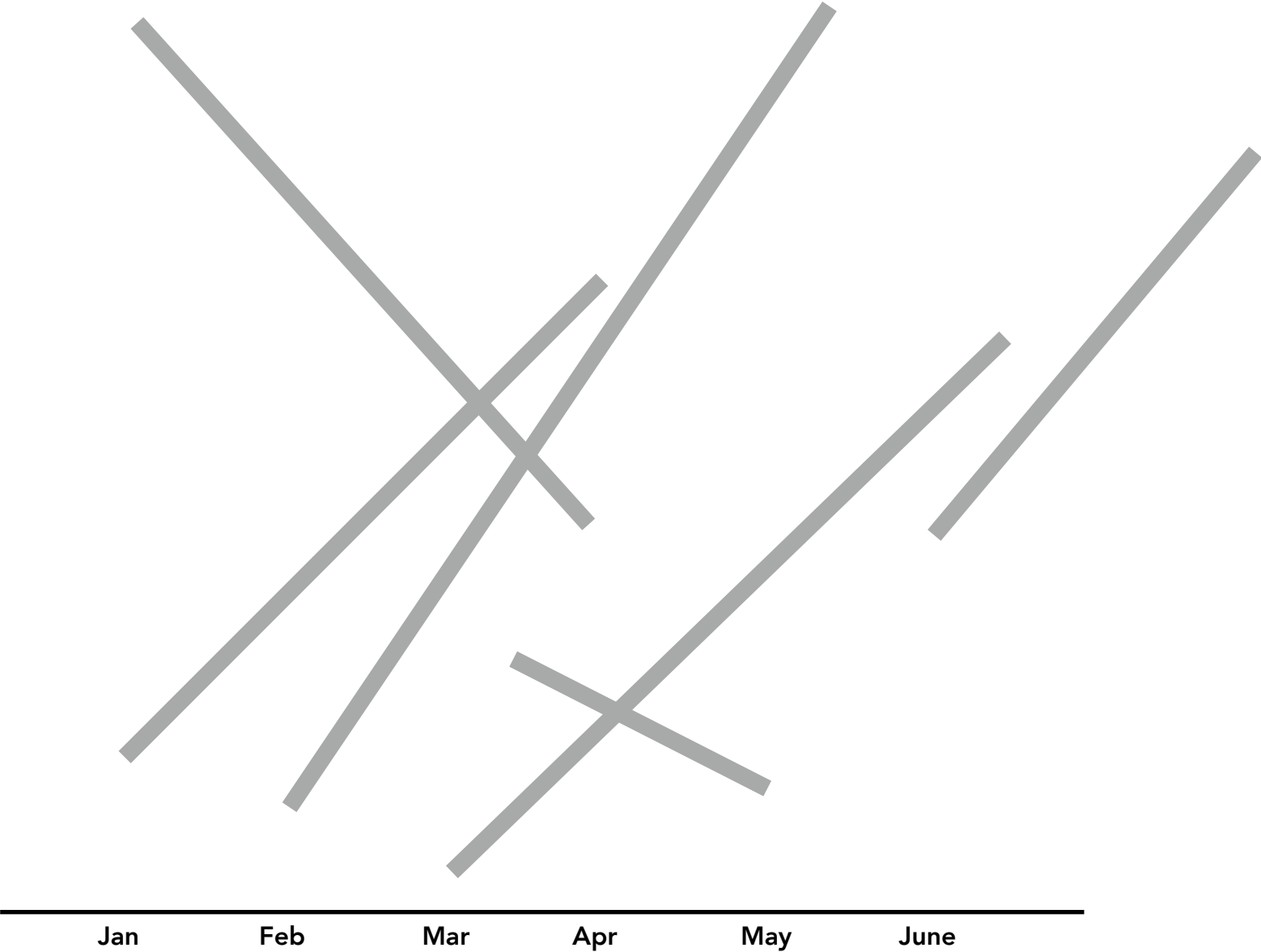


Fig. 11. Incorrect use of a bar chart for the *Nation* relation. The lengths of the bars suggest an ordering on the vertical axis, as if the USA cars were longer or better than the other cars, which is not true for the *Nation* relation.

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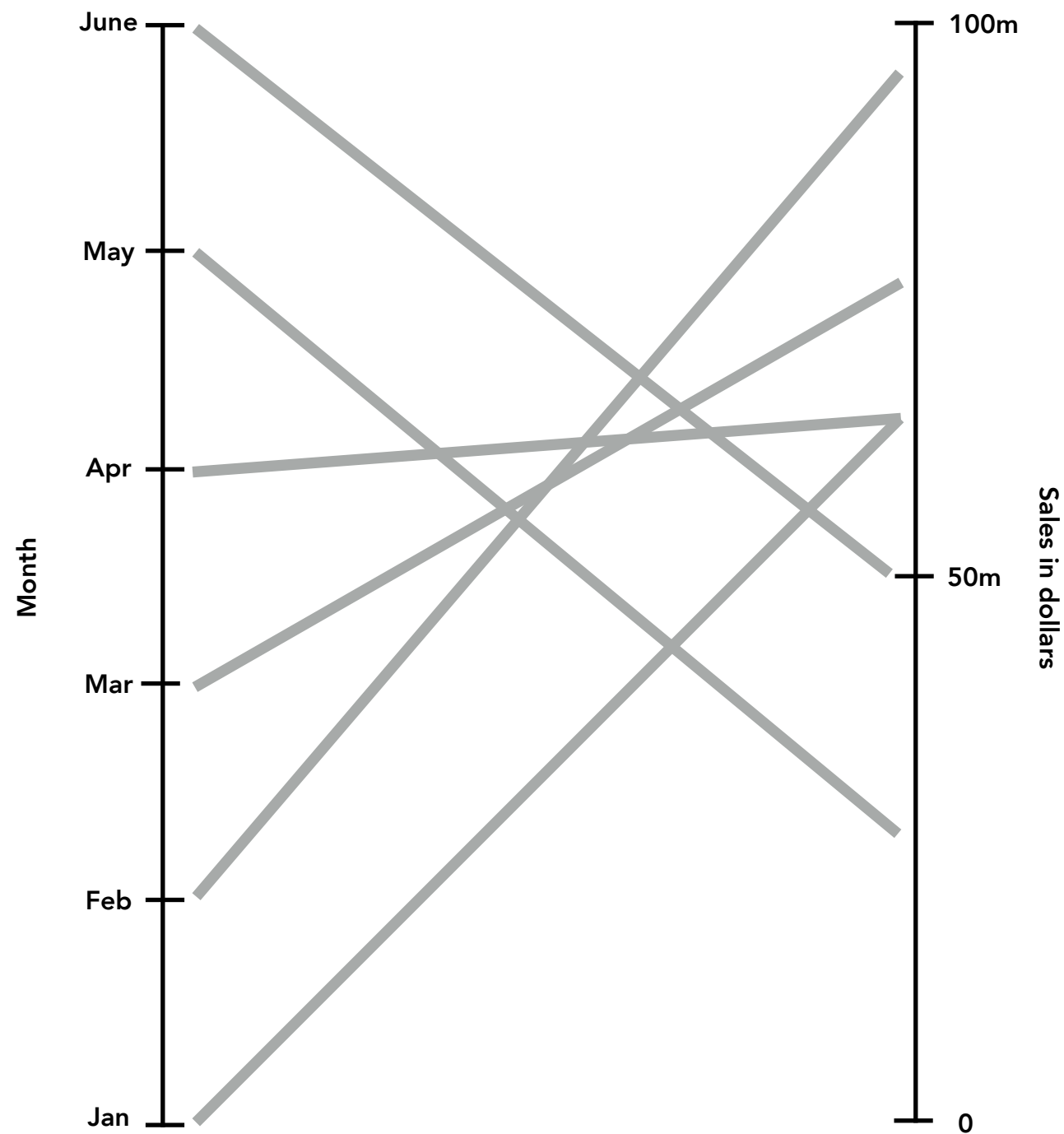


**Month** is represented by the ordinal rank of the x-position of the **bottom of the bar**.

**Sales** are represented by the length of the bar.

Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 2: slope graph



Sales by Month, ACME Corp., 2025

**Expressively equivalent** to standard bar charts.

**Perfectly uniform and sign-dependent**, so fully iconic.

Yet **difficult** and **unintuitive**.

## In search of...

- A theory of cognitive resonance that...
  - Prioritizes **hard factor** system design over **soft factor** graphic design.
  - **Avoids primitive** representational **schemas** and **scales**.
  - **Explains** why diverse **dissonant** diagrams **are difficult to use**.
  - **Explains** why diverse **resonant** diagrams **are easy to use**.

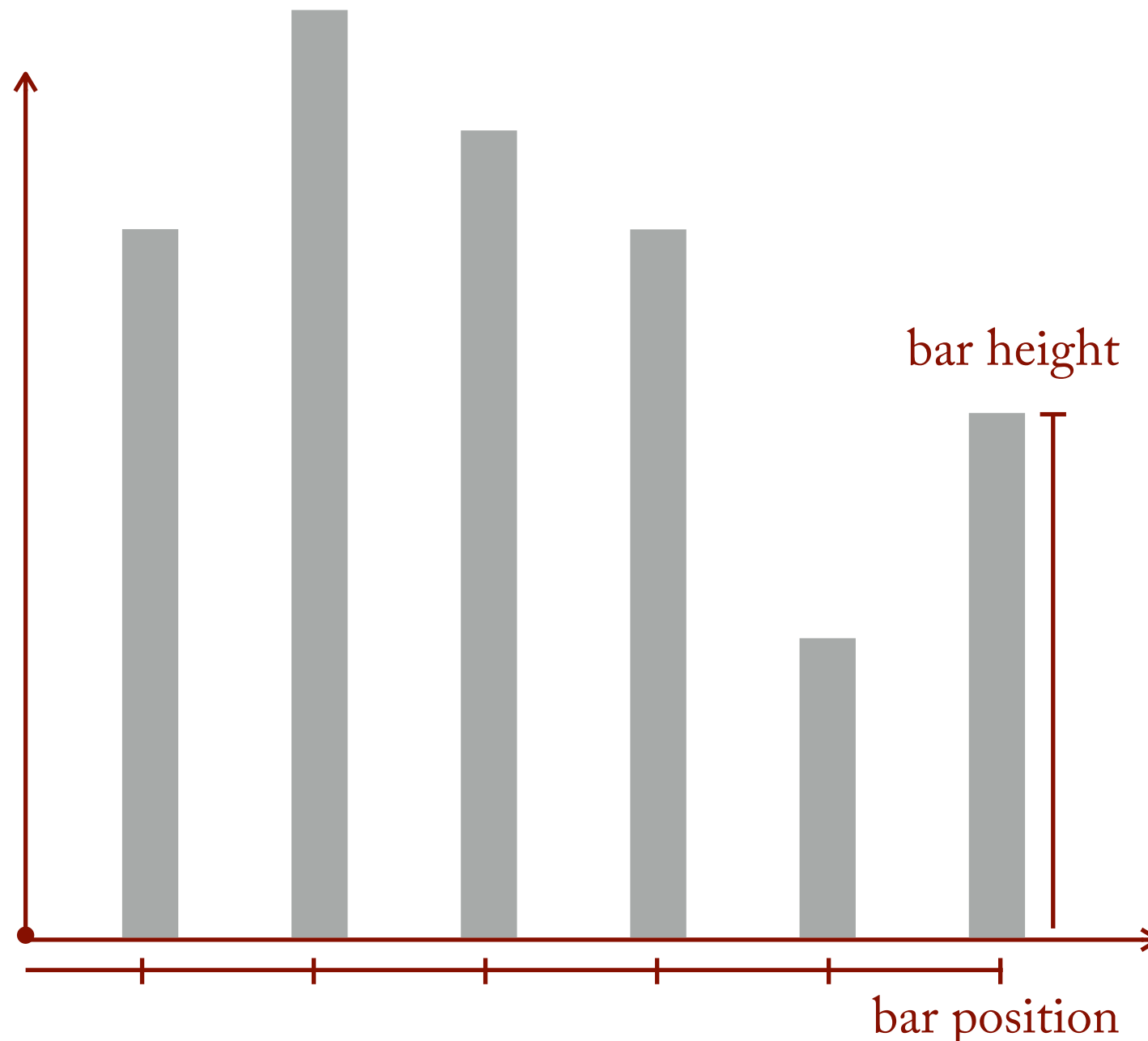
# Visual strategies

## Sensible to the eye

- “These circles... are extremely commodious for facilitating our reflection on this subject, and for unfolding all the boasted mysteries of logic, which that art finds it so difficult to explain; whereas by means of these signs the whole is rendered sensible to the eye.”  
— Euler (1768) *Letters to a German Princess*
- But what is **sensibility to the eye**?
- **Model 1:** the use of visual perception in **reading text**.
  - Visual system: **shape perception and categorization**.
- **Model 2:** the use of visual perception in interpreting pictures.
  - Visual system: **full spatial perception** (depth, color, etc.) for static scenes.
- Neither model fits the case of diagrams.
  - Not text categorization; not visual space. Then what?

# Common wisdom: vision as a measurement device

- In diagram interpretation, the visual system acts as a **measurement device**.  
(Bertin, Tversky, Mackinlay, Pinker)
- Vision is fine-tuned for measurements of **magnitudes, orders, and comparative relations**.
- Interface with the **analogue magnitude system** in core cognition.



Bertin, J. (1967/1983). *Semiology of Graphics*,  
Mackinlay, J. (1986). Automating the design of  
graphical presentations of relational information.  
Pinker, S. (1990). A theory of graph  
comprehension.  
Tversky, B. (2011). Visualizing thought.

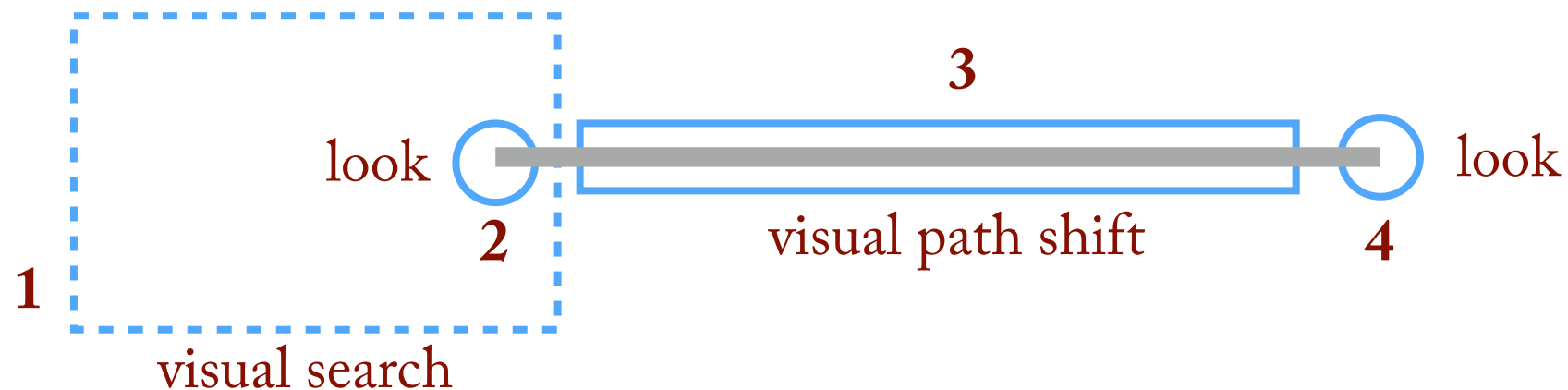
# Strategies for visual measurement

- We do not (and cannot) extract all information from a diagram in a single glance.
- **Embodied perspective:** to read a diagram, we must inspect it with the limited perceptual capacities ready to hand. Compare the visual agent to a Turing Machine (or Snakes and Ladders player).
- Viewers must employ **visual strategies** to extract information from a diagram. These are techniques for how to **read** a diagram.
- A **visual strategy** is a planned sequence of visual actions designed to perform the measurements necessary to answer a query about the data.
- Visual strategies are part of diagram literacy.
- **Key hypothesis:** resonant diagrams are those which support efficient visual strategies for common queries.

# Looks, shifts, and searches

- Visual measurements are based on individual **looks**. A look is defined by an allocation of attention to a part of a visual object and a visual measurement of its features.
- Looks are linked together by **shifts**:
  - Linear shifts follow a **visual path**, like a line (cheap)
  - General **visual search** ranges over 2D space (expensive).
- Visual measurements can be stored in **visual memory** (cheap) or **working memory** (expensive).
- In resonant diagram systems, information can be extracted by an efficient strategy of looks, shifts, and memory storage.

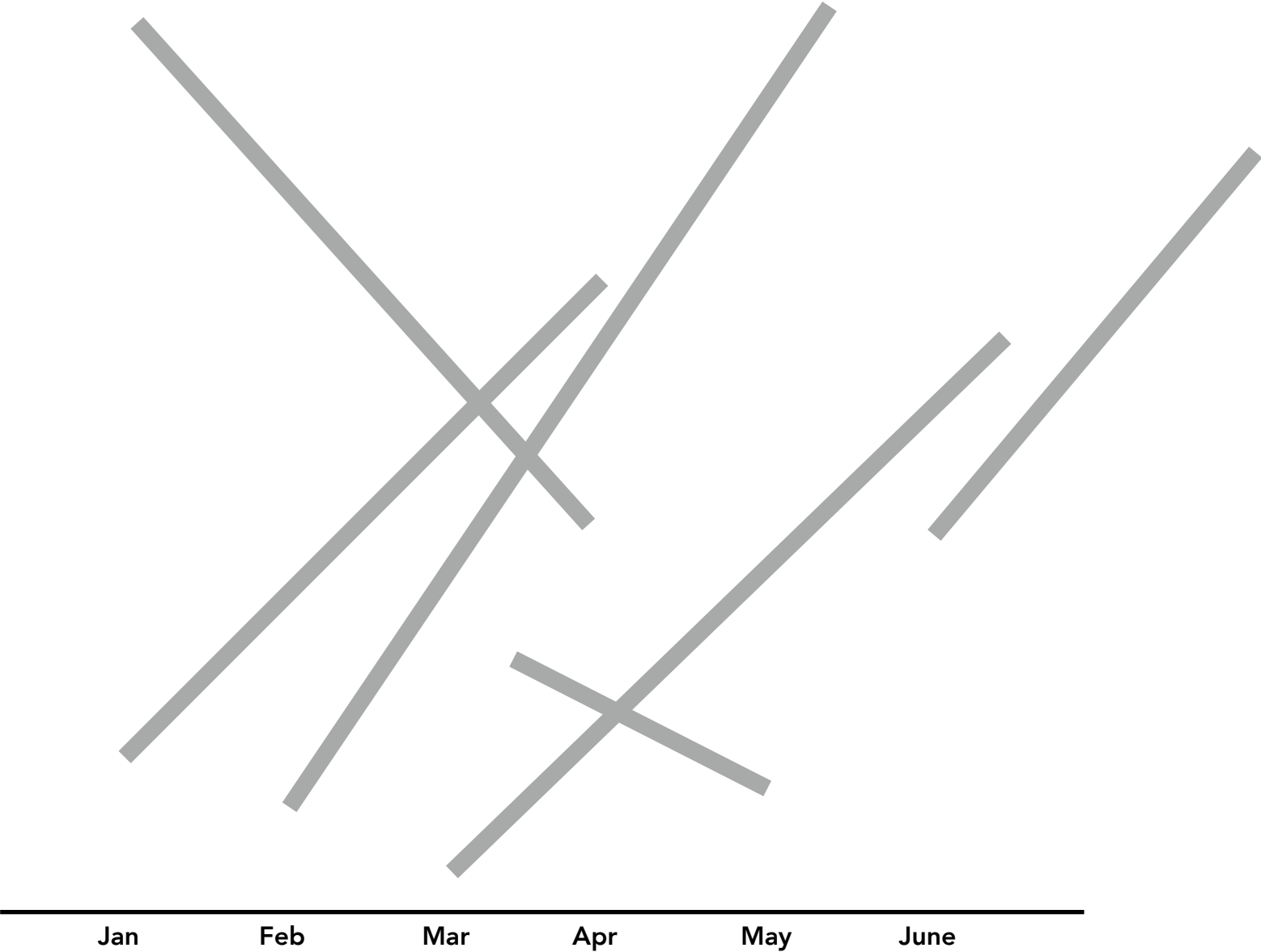
Query: how long is the bar?



## Queries

- Visual strategies have as their **end goal** resolving some **query** about the data.
- There are any number of possible data queries, and a successful diagram system must support many different common queries.
- The most basic queries are **data-point queries**: queries about the intersection of the two data dimensions.
- The **standard form** for a data-point query specifies its location on one data dimension, and asks about the other.
- So for any given data point in the sales by month chart, we can ask:
  - **Q1.** In month X, what were the sales?
  - **Q2.** For sales Y, in what month did they occur?

# Dissonant diagrams, Case 1: scattered bars

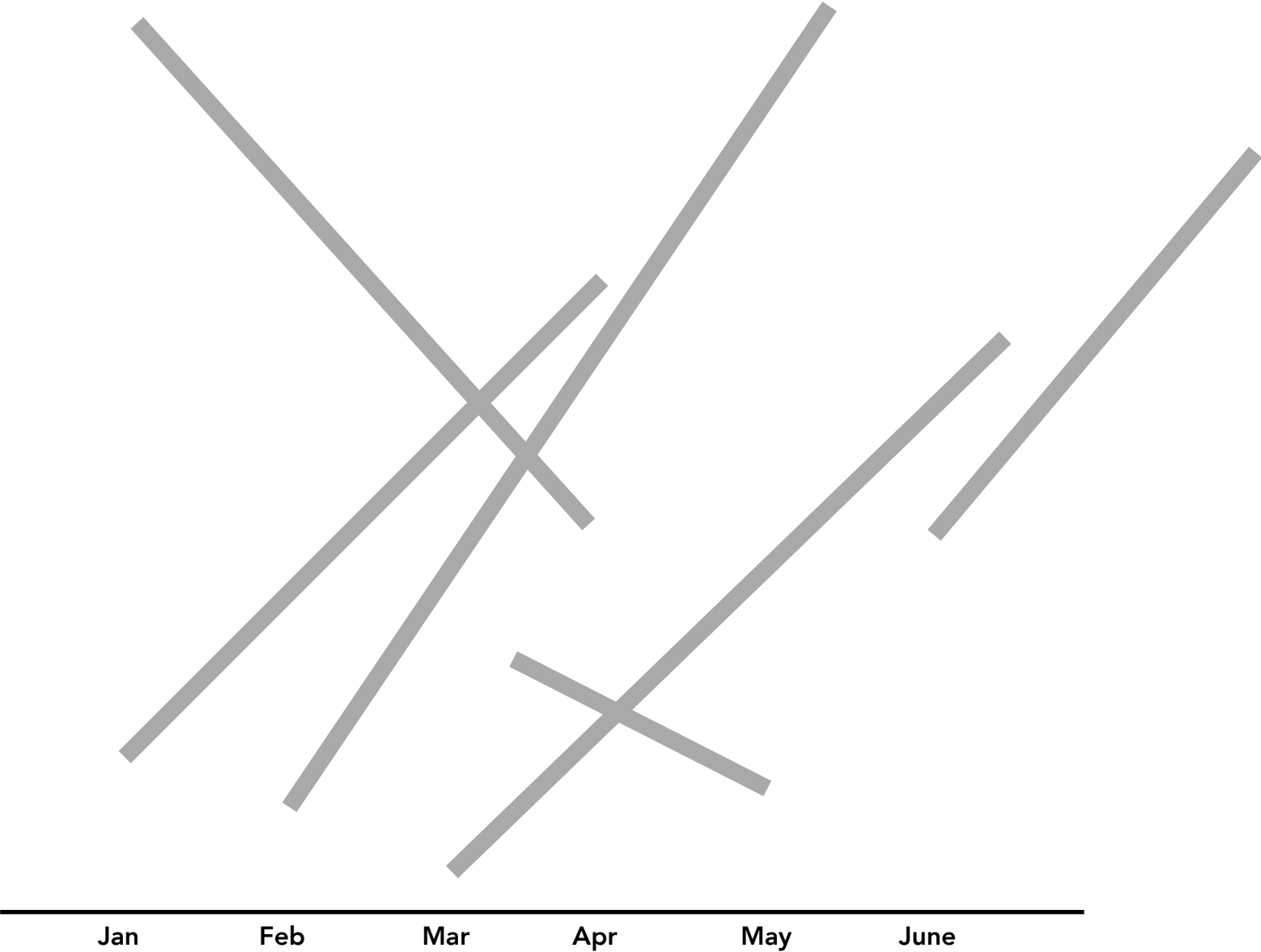


**Month** is represented by the ordinal rank of the x-position of the **bottom of the bar**.

**Sales** are represented by the length of the bar.

Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 1: scattered bars



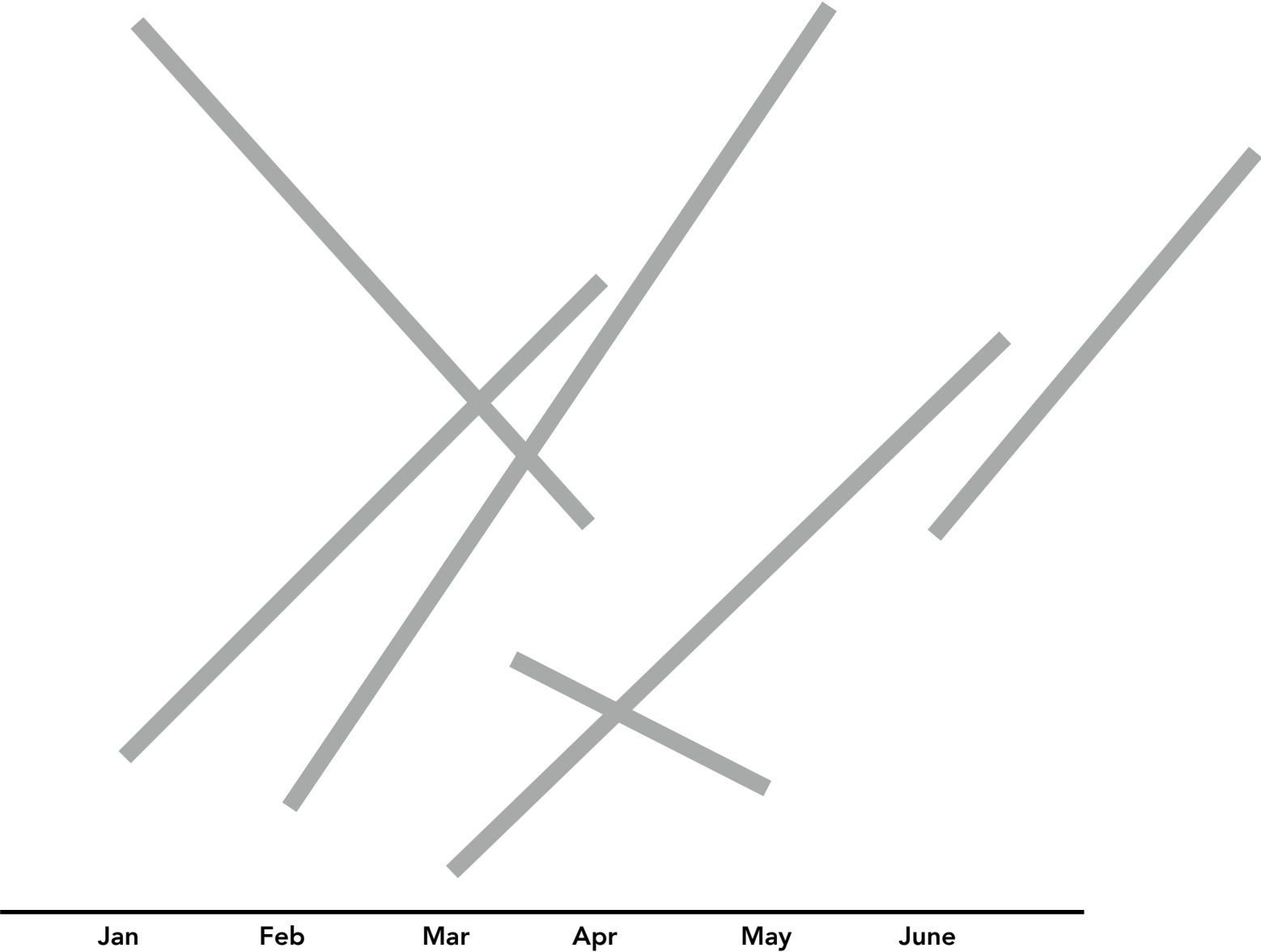
Sales by Month, ACME Corp., 2025

**Expressively equivalent** to standard bar charts.

**Perfectly homomorphic/ iconic** in its semantics.

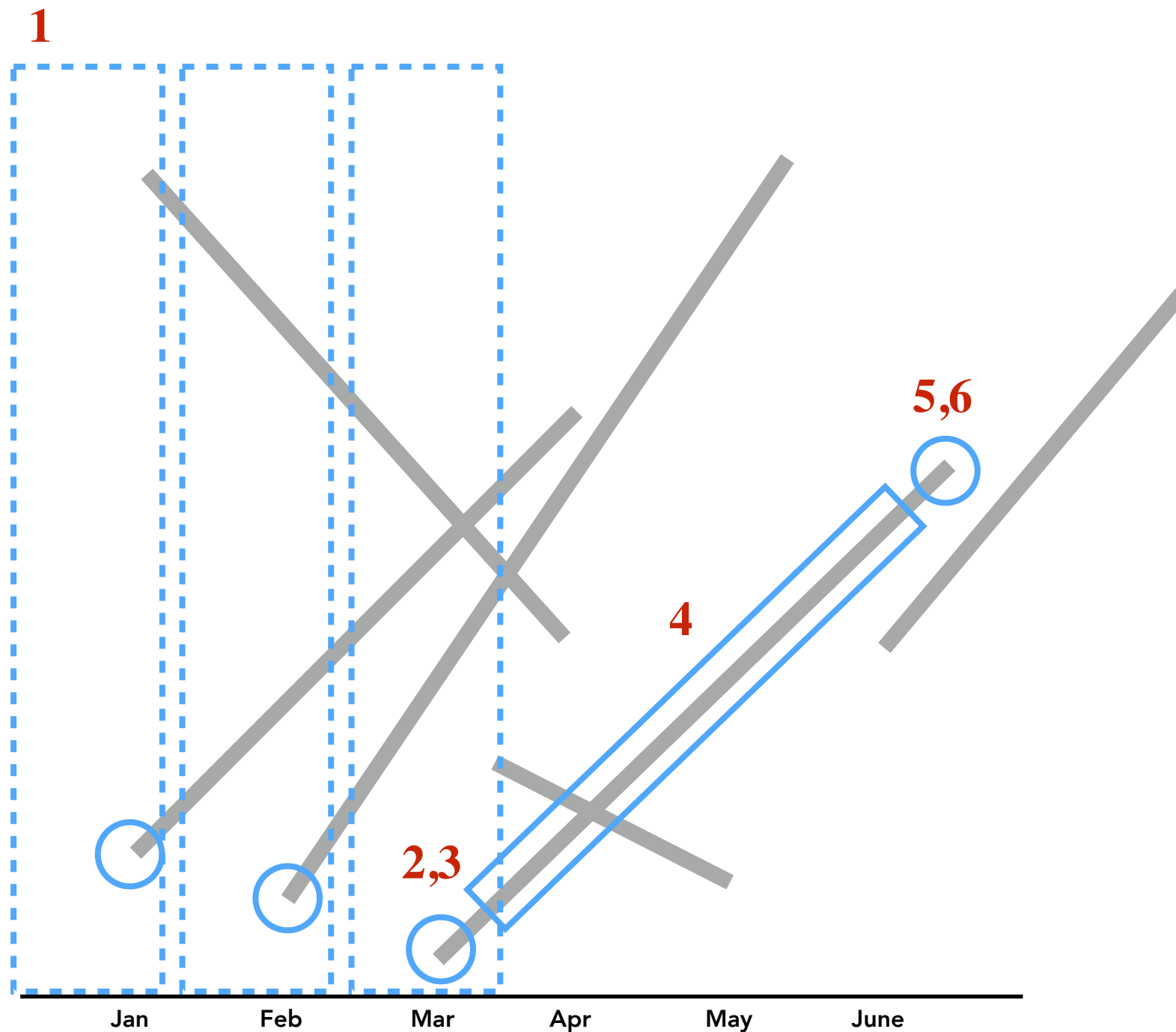
Yet **difficult** and **unintuitive**.

Query: what were sales in the third month?



Sales by Month, ACME Corp., 2025

**Query: what were sales in the third month?**



Sales by Month, ACME Corp., 2025

**Step 1:** iterative 2D search for third bar low points

**Step 2:** store “3” in WM.

**Step 3:** measure xy-position of current bar-end. Store in WM.

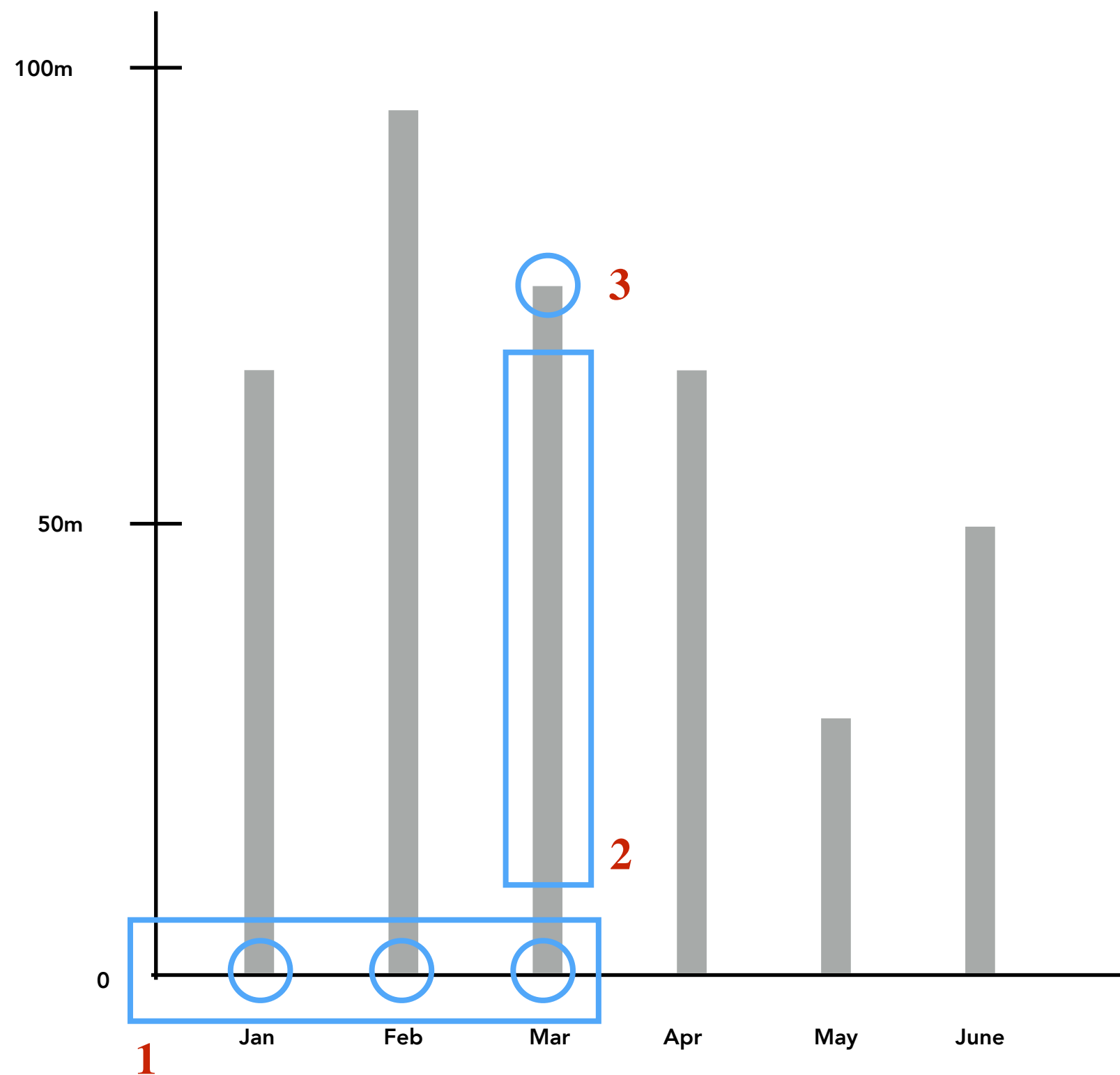
**Step 4:** traverse line.

**Step 5:** measure xy-position of current bar end.

**Step 6:** Compute bar length.

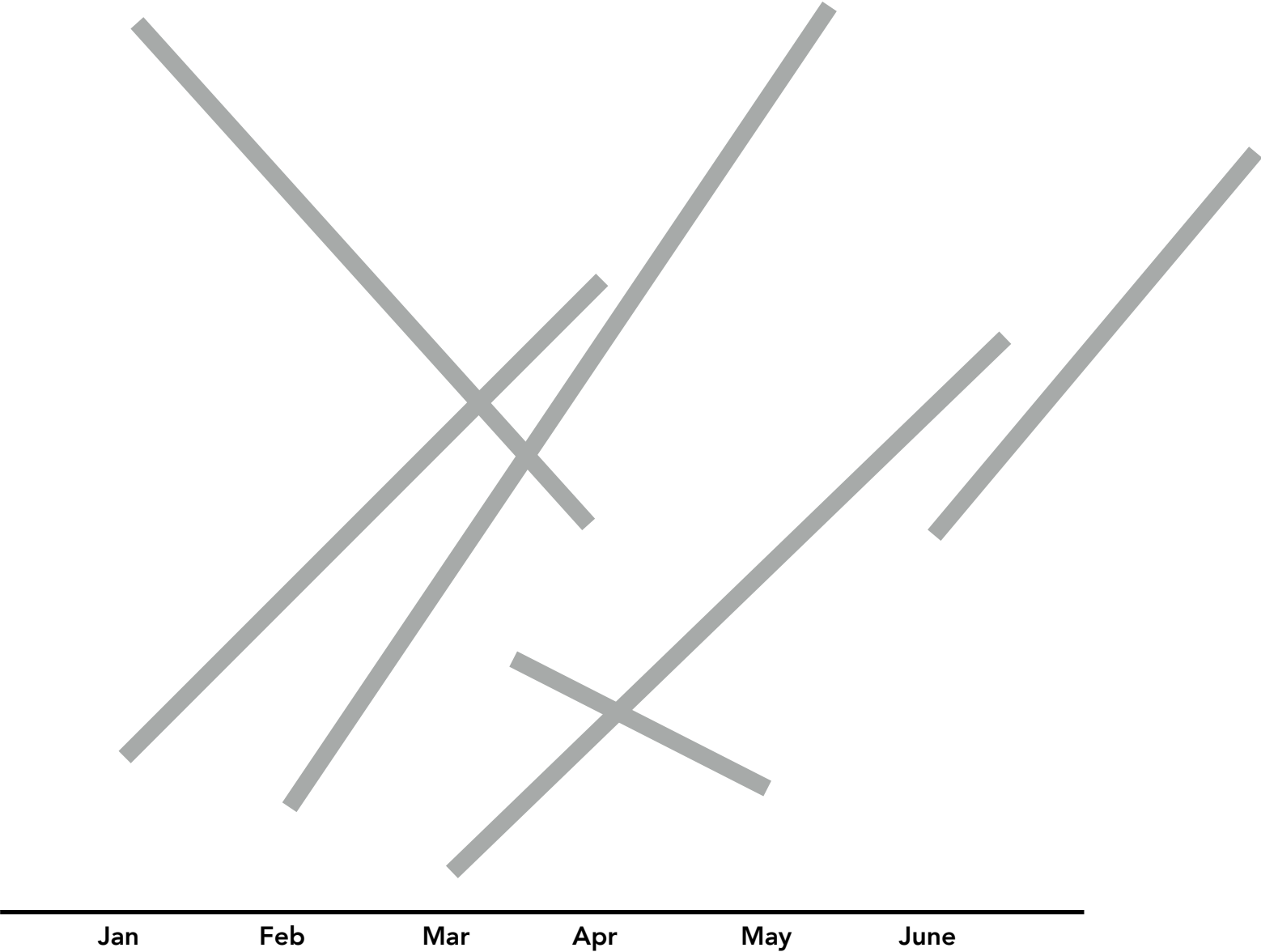
**Step 7:** Answer query.

Query: what were sales in the third month?



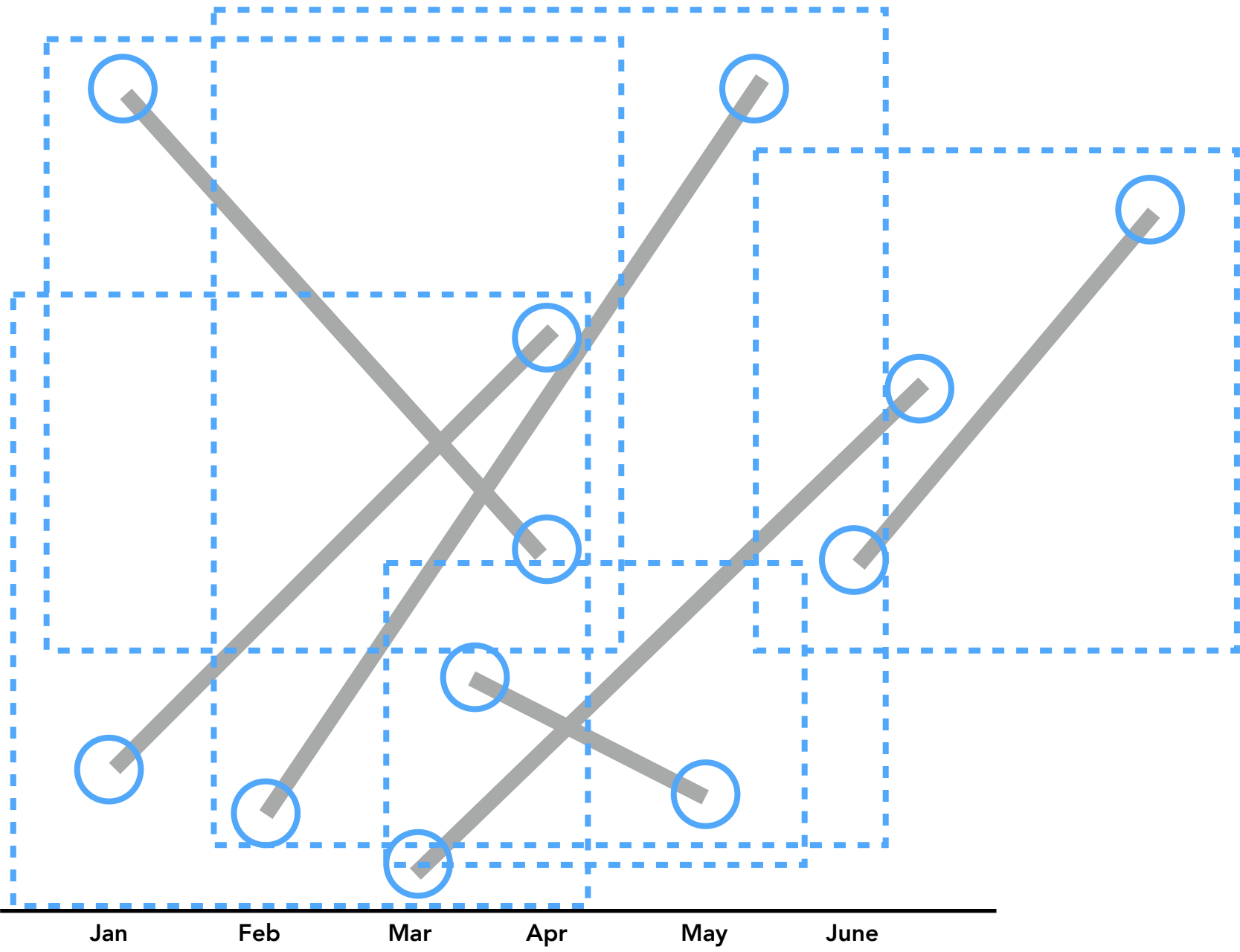
Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



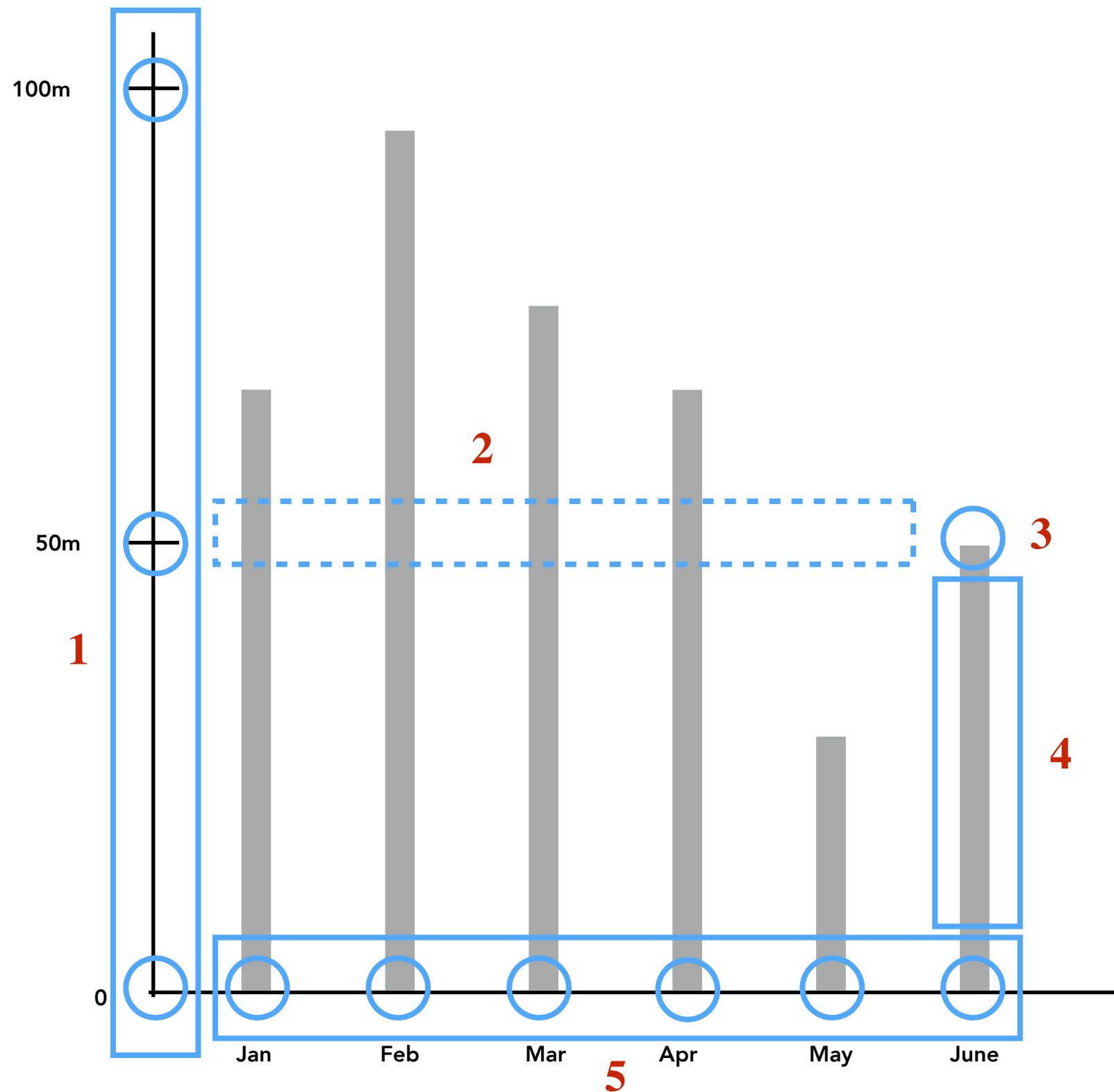
Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



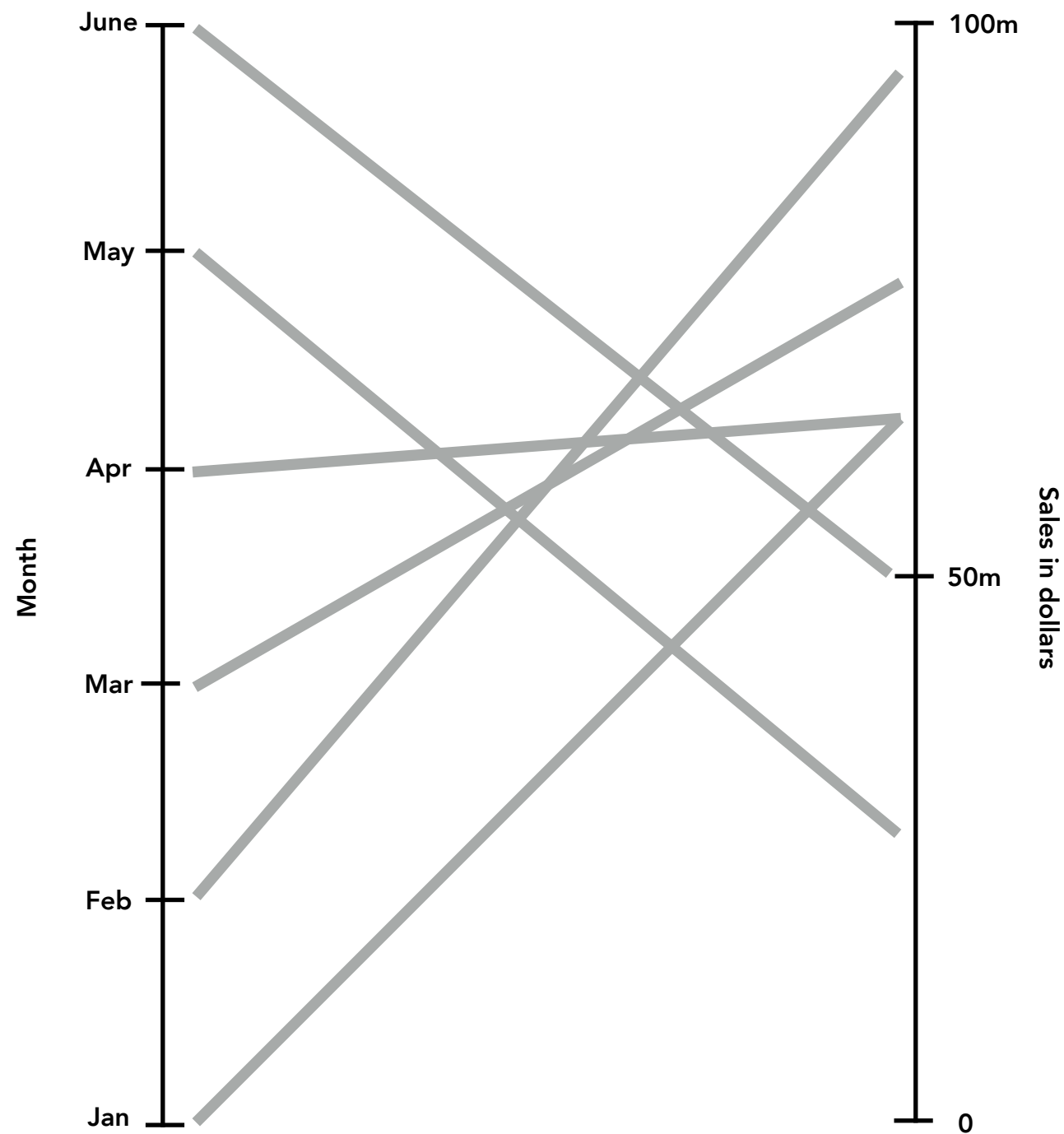
Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 2: slope graph



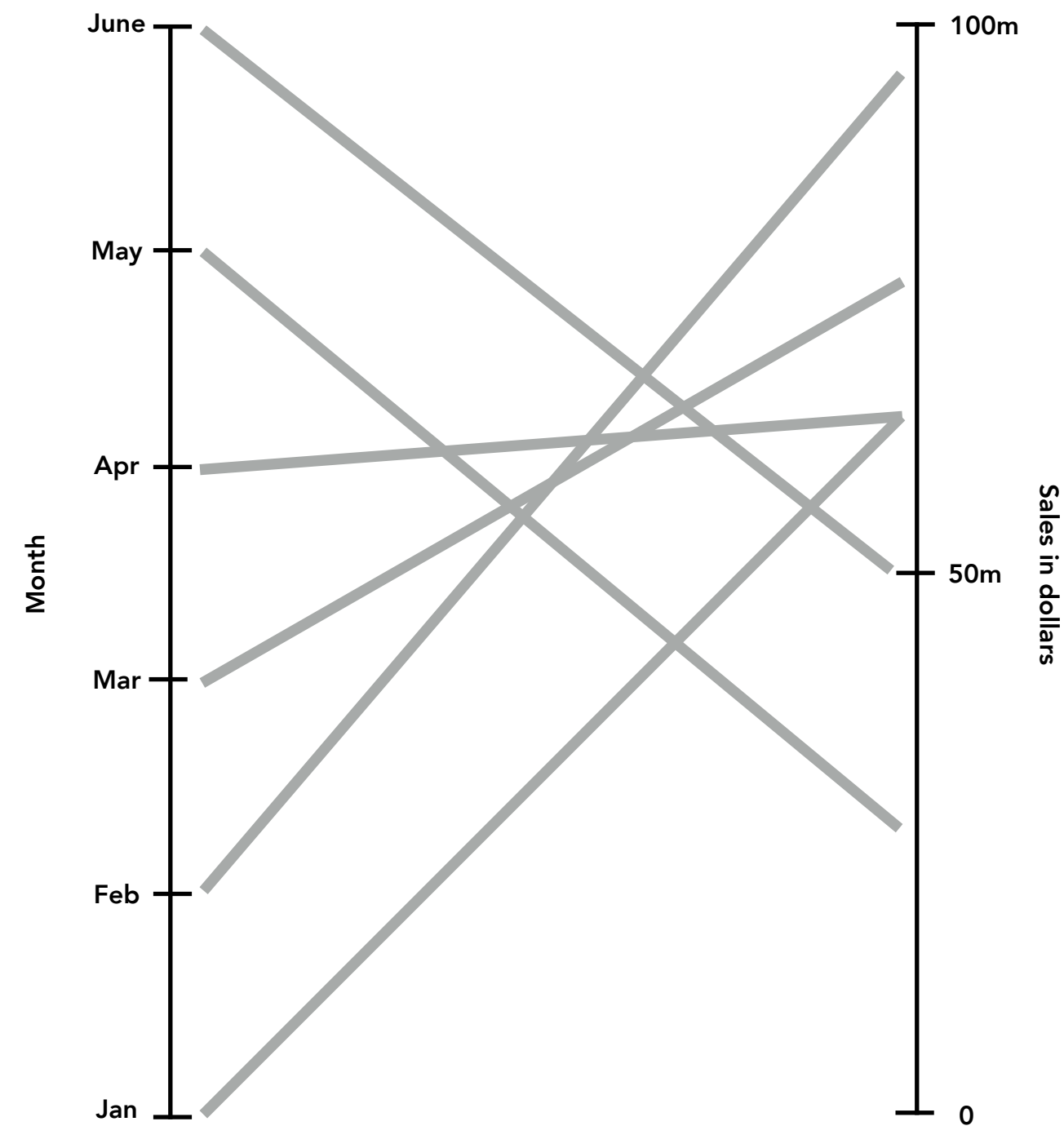
Sales by Month, ACME Corp., 2025

**Expressively equivalent** to standard bar charts.

Perfectly homomorphic/  
iconic in its semantics.

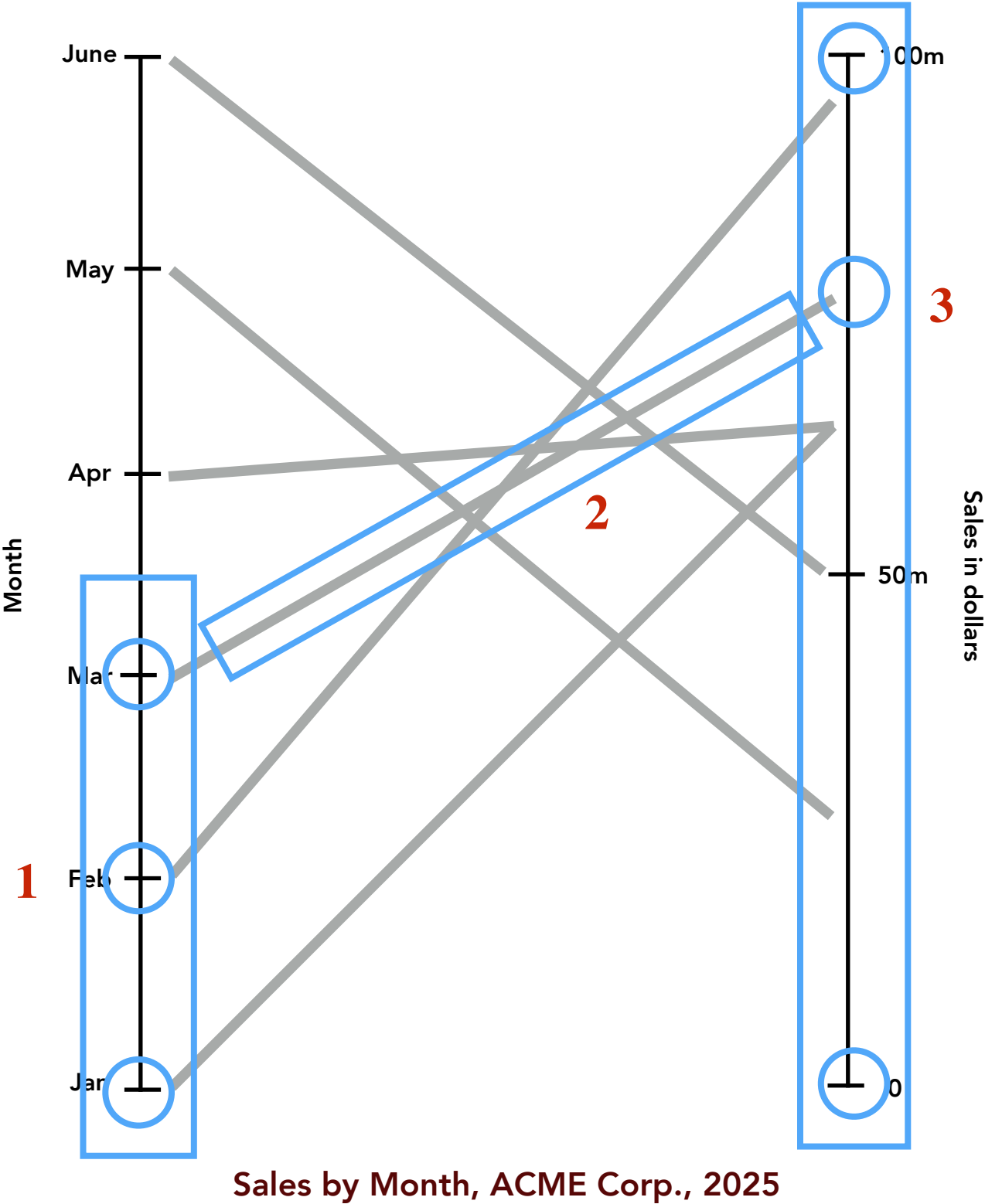
Yet **difficult** and **unintuitive**.

Query: what were sales in the third month?



Sales by Month, ACME Corp., 2025

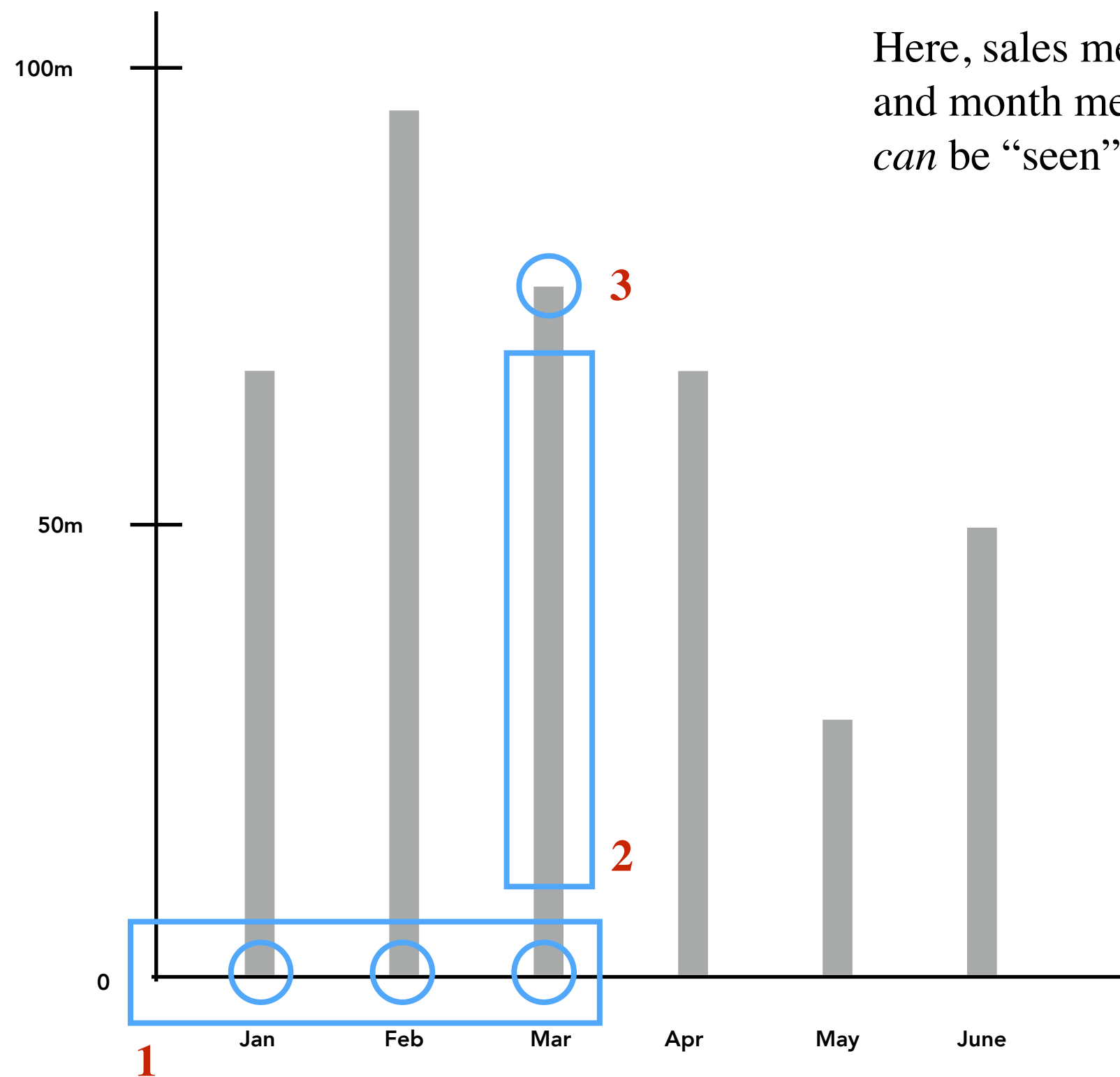
# Query: what were sales in the third month?



Reliance on working memory in the transition from 1 to 3.

Sales measurements and month measurements cannot be “seen” at once.

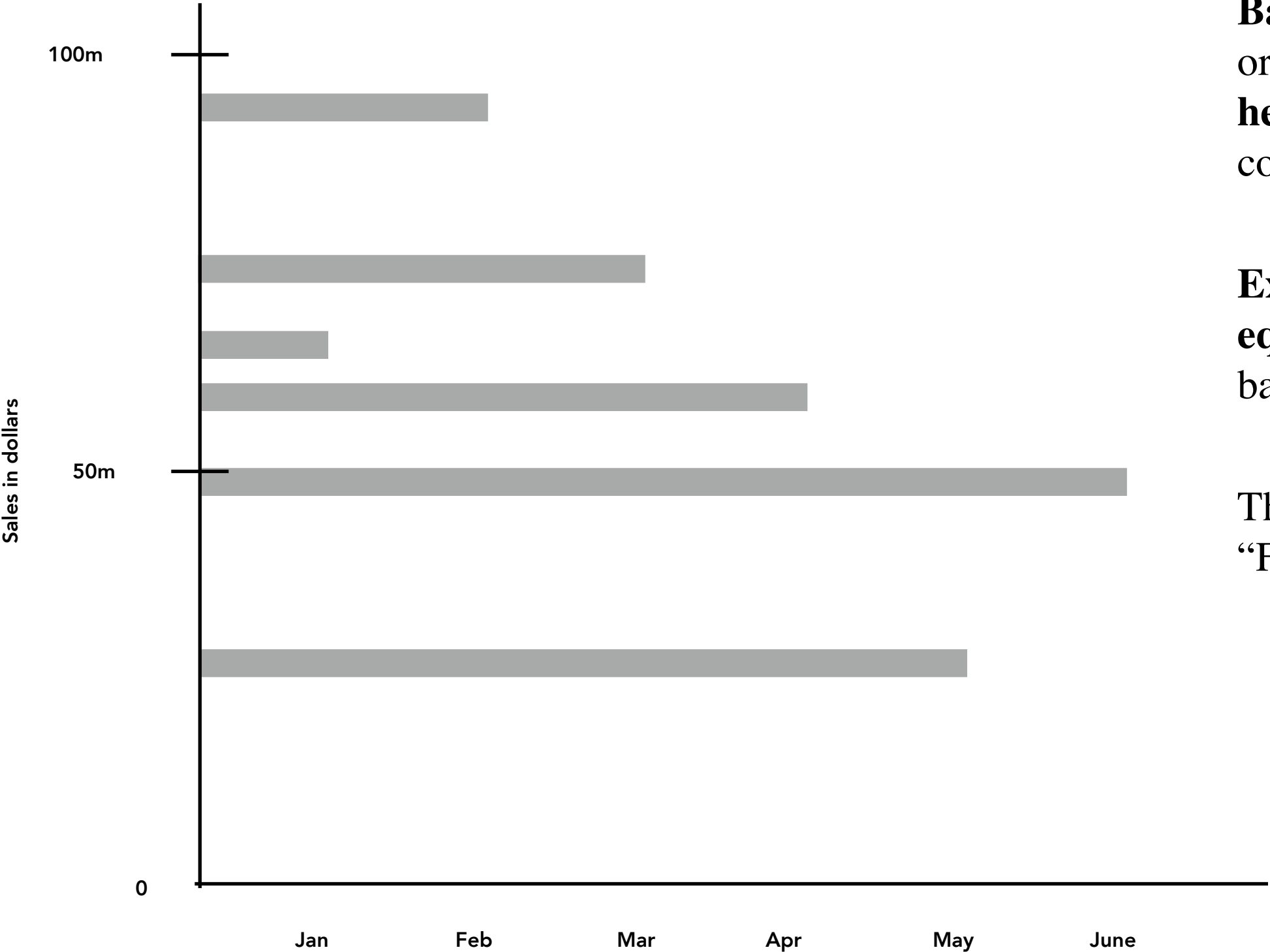
Query: what were sales in the third month?



Here, sales measurements and month measurements *can* be “seen” at once.

Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 3: ordinal bars



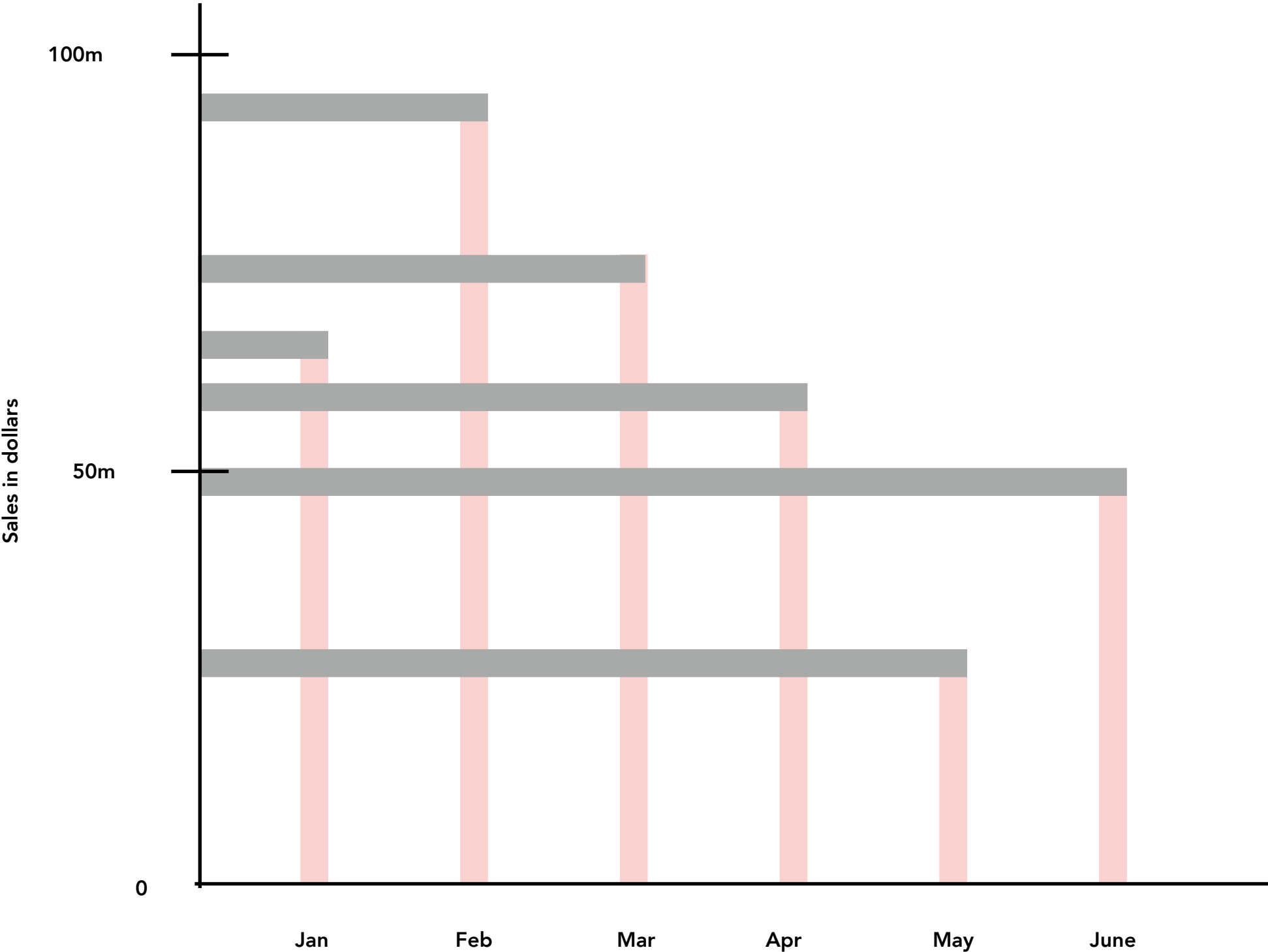
**Bar length** represents ordinal X-position. **Bar height** represents continuous Y-value.

**Expressively equivalent** to standard bar charts.

The hardest of the “Frege cases.”

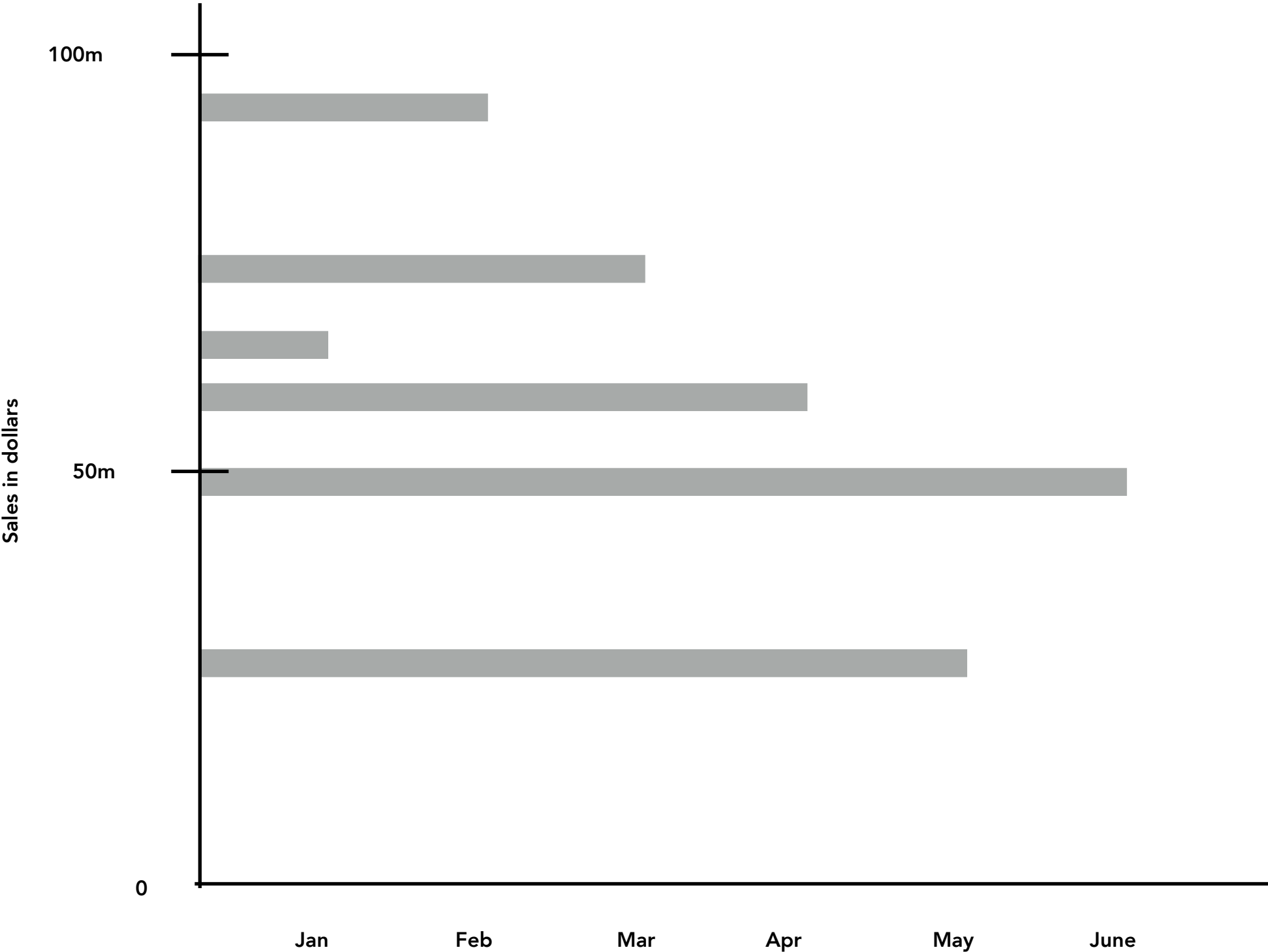
Sales by Month, ACME Corp., 2025

# Dissonant diagrams, Case 3: ordinal bars



Sales by Month, ACME Corp., 2025

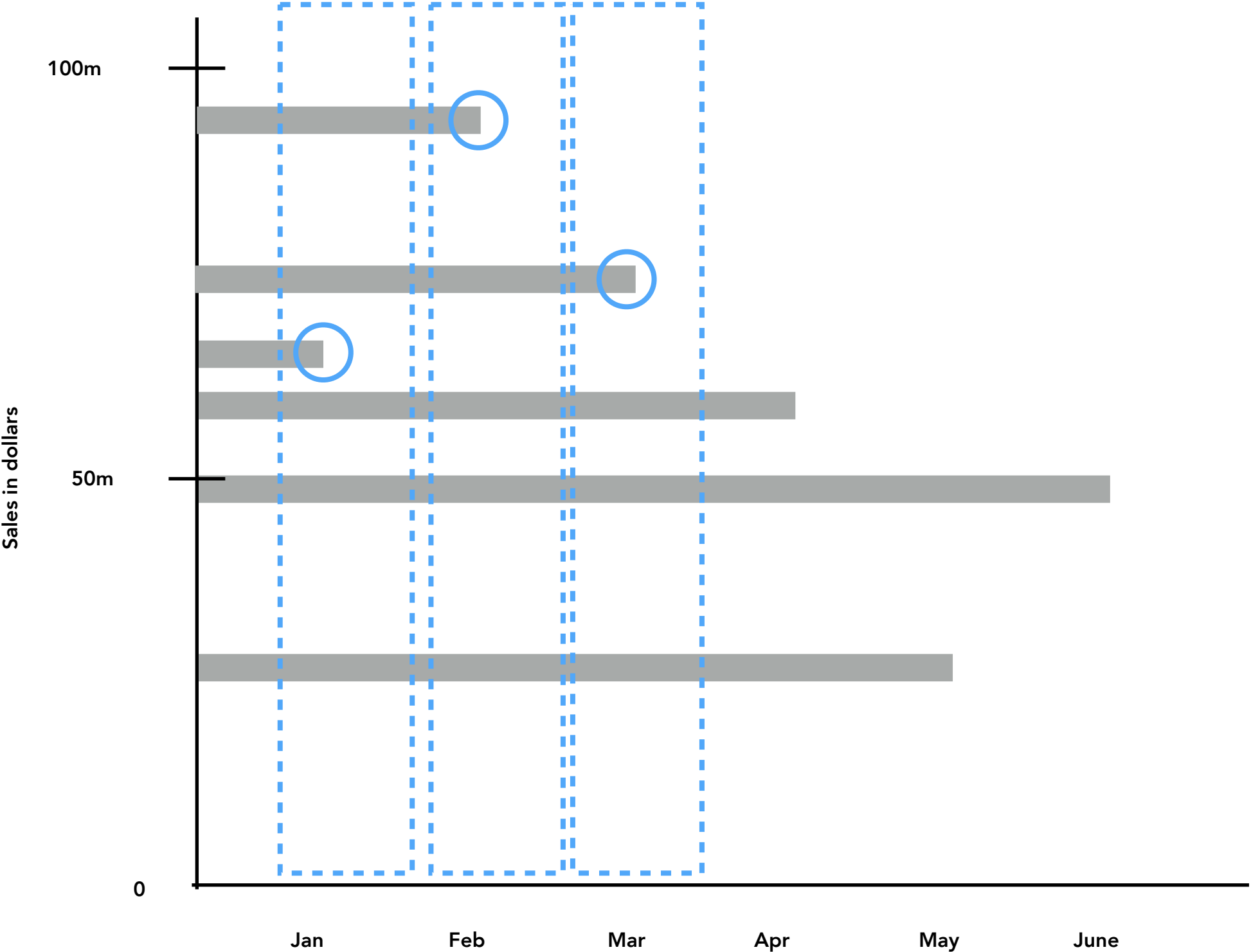
Query: what were sales in the third month?



Sales by Month, ACME Corp., 2025

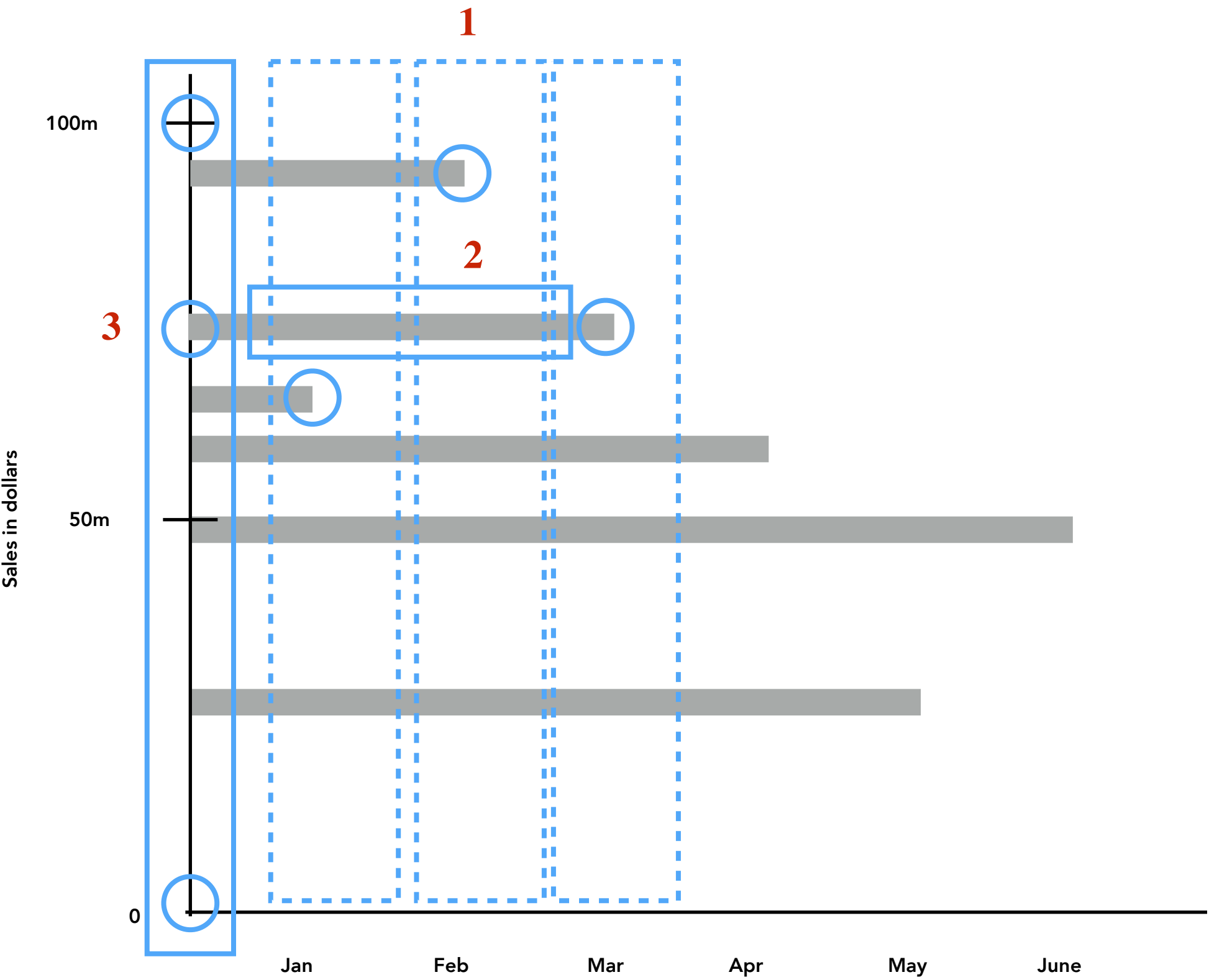
Query: what were sales in the third month?

1



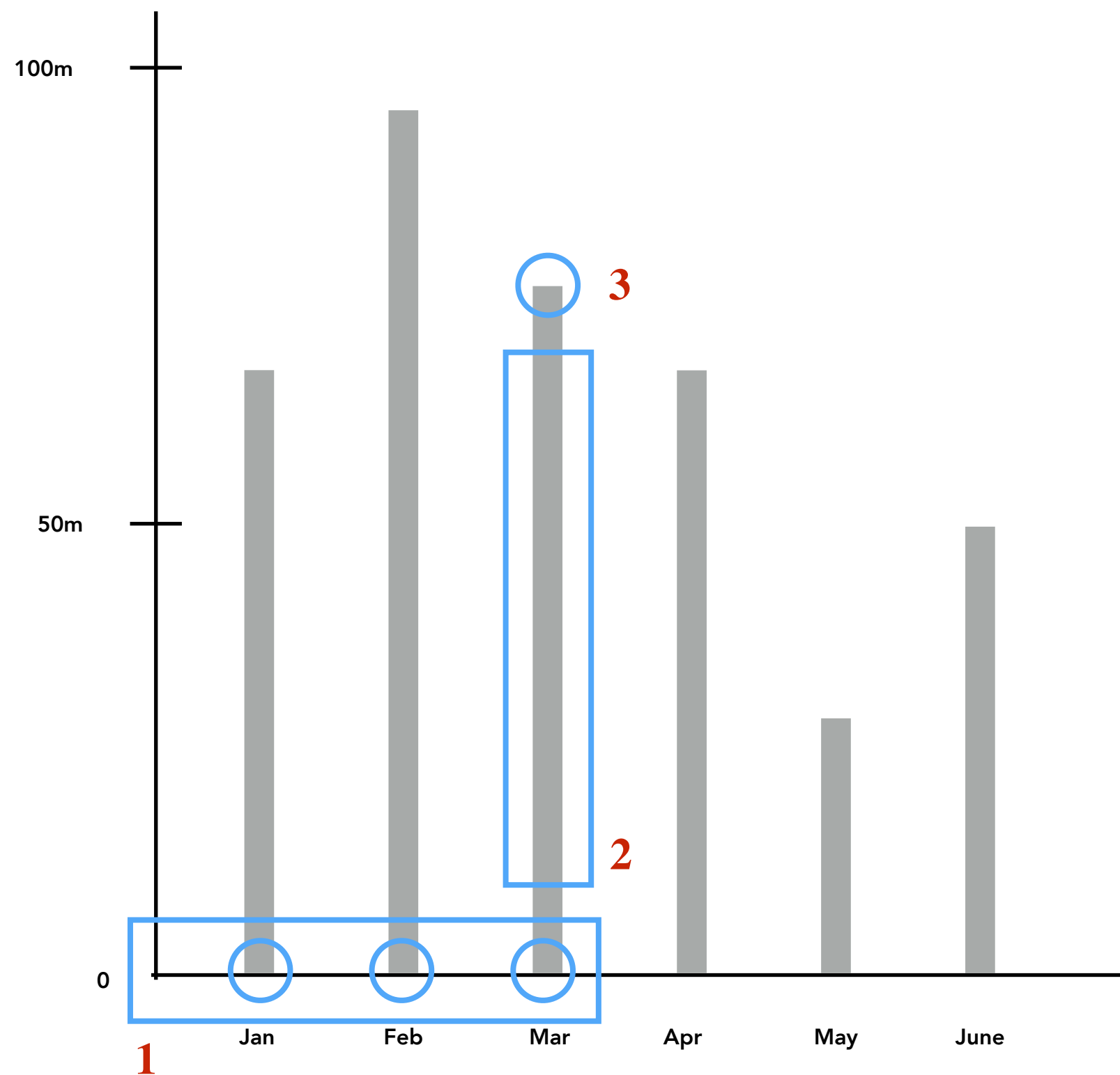
Sales by Month, ACME Corp., 2025

Query: what were sales in the third month?



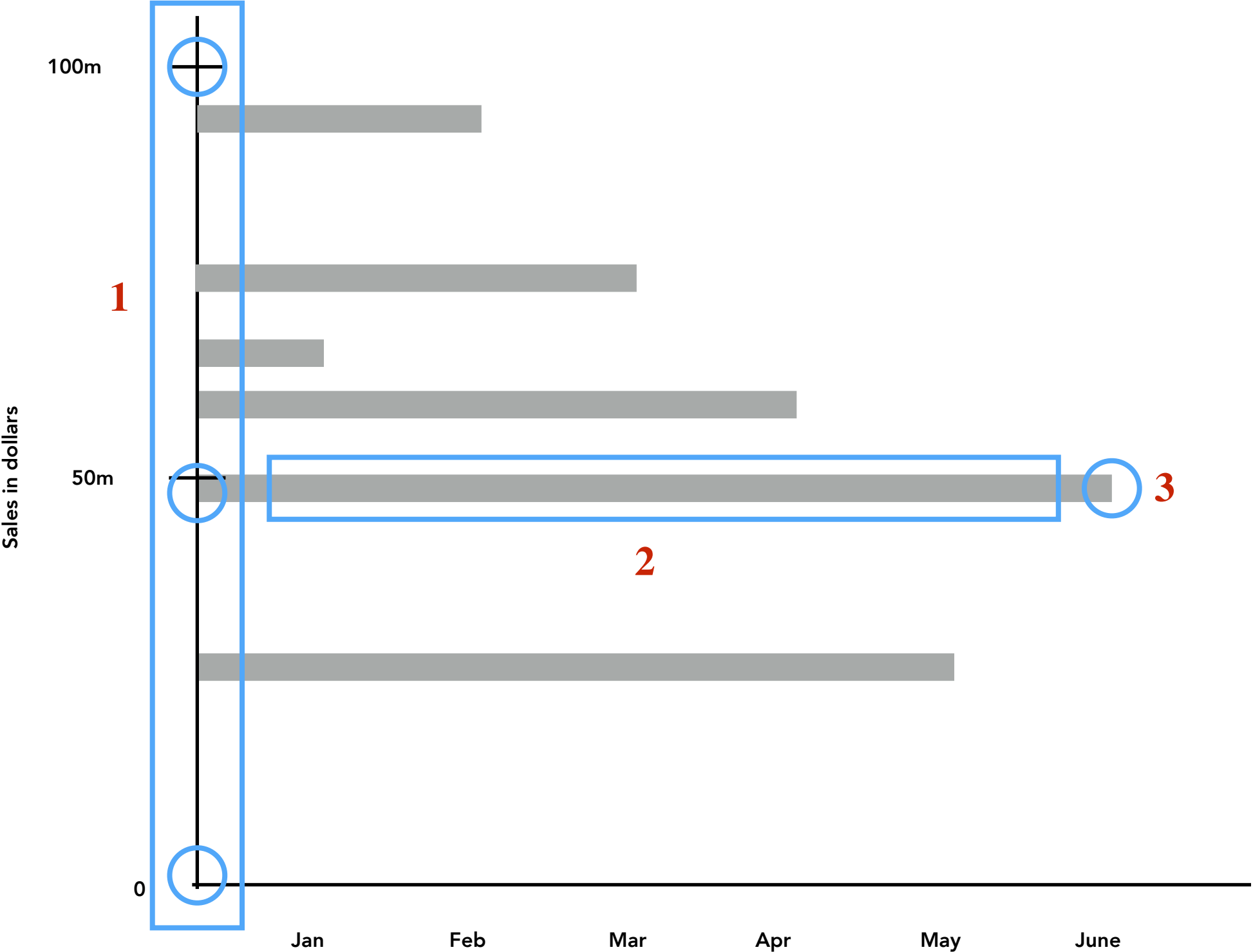
Sales by Month, ACME Corp., 2025

Query: what were sales in the third month?



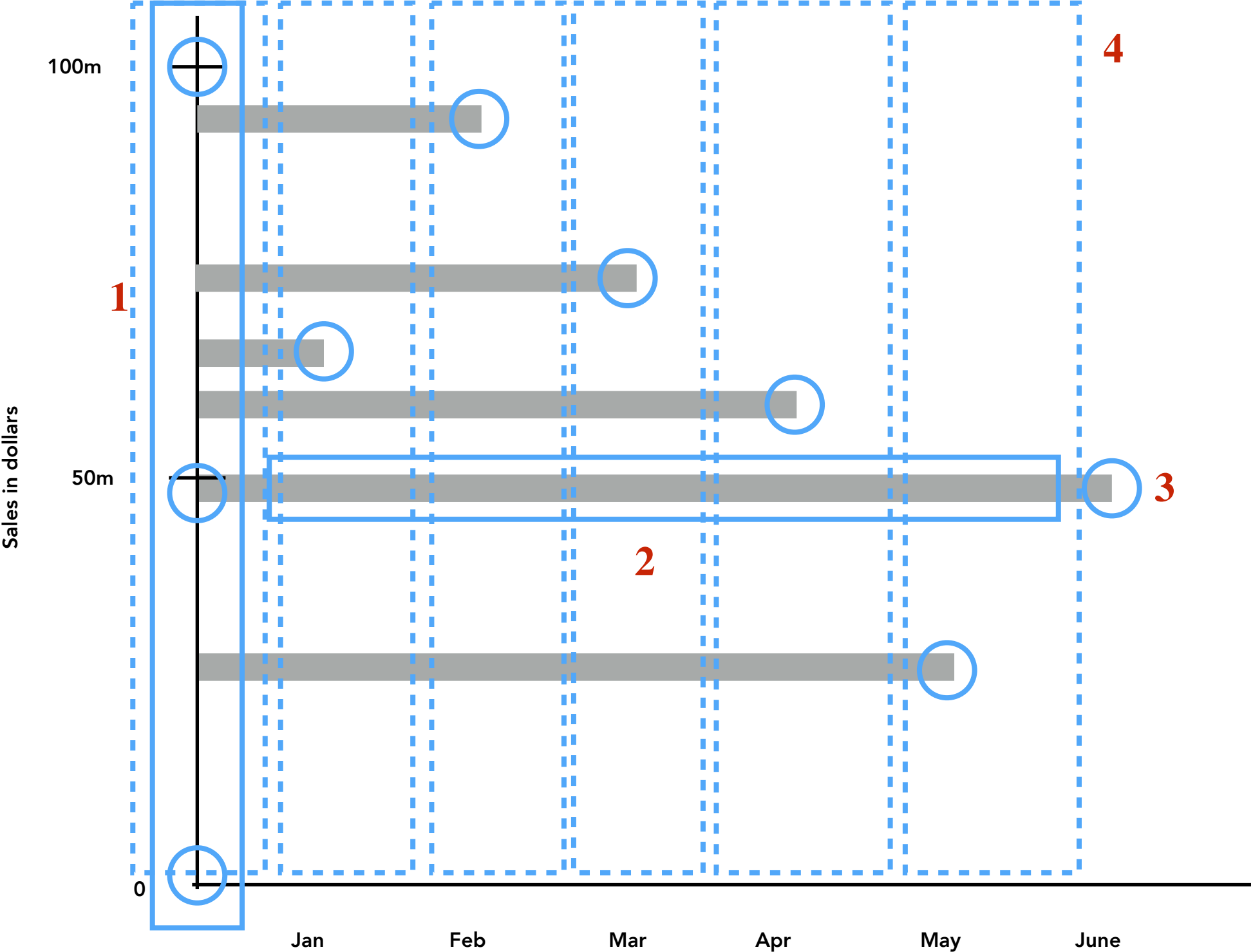
Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



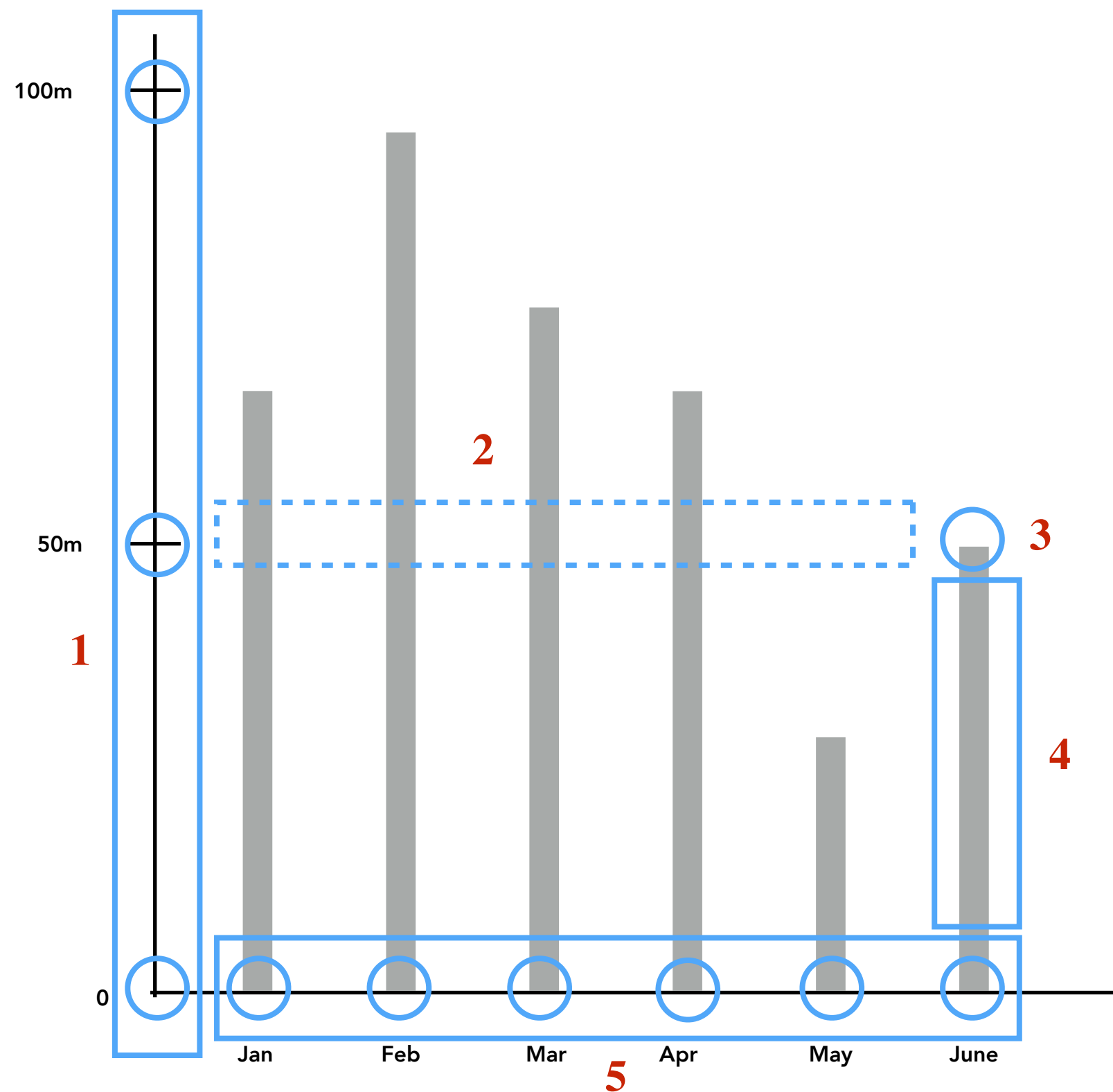
Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



Sales by Month, ACME Corp., 2025

Query: in what month were sales approx. 50m?



Sales by Month, ACME Corp., 2025

## The moral of the ordinal bars

- **A “Frege case” for diagrams:** same data expressed, same *visual information* provided. Different organizing lines. Different degrees of resonance.
- **Visual strategy explanation:** the ordinal bar graph makes ordinal information difficult to extract.
- **Asymmetry-based explanation:** sales by month is a *function* from *months* to *sales*. The diagram should reflect this dependency.
- **Redundancy-based explanation:** since sales/month is a *total* function, the ordinal frame of reference already encodes the order of months. By contrast the continuous Y-dimension doesn't encode the order of values.
- Finding the right fit between the method of extraction and the structure of the information itself.

## Summary

- **Diagrams** are non-pictorial iconic representations, scaffolded over very simple semantic rules.
- Why do some diagram systems **flourish** after their introduction, while others fade away?
- Flourishing must be explained in part by an account of the **cognitive capacities** that diagrams recruit.
- Some diagram systems are **cognitively resonant** due to the way they enlist our ability to **visually navigate and measure** a diagram en route to answering data queries.
- Cognitively resonant diagrams support **efficient visual strategies**.



Thank you